

REVIEW

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# Virtual reality in education: global trends, challenges, and impacts—game changer or passing trend?

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## Abstract

Virtual reality (VR) has emerged as a transformative technology in education, enabling immersive simulations that foster student engagement and improve knowledge retention. This study presents a bibliometric analysis of global trends, research patterns, and challenges related to the integration of VR in educational contexts. Data were collected from the Science Citation Index Expanded (SCIE), Social Sciences Citation Index (SSCI), and Emerging Sources Citation Index (ESCI) within the Web of Science Core Collection, including only articles also indexed in Scopus to ensure high-quality and comprehensive coverage. Of the 1220 articles published between 2010 and 2024, 97 met the inclusion criteria based on relevance, accessibility, and methodological rigor. We performed keyword analysis, citation analysis, and thematic mapping using RStudio and the Bibliometrix package. The findings indicate that the United States, China, and the United Kingdom are the most active contributors to VR research in education. VR has been applied across diverse learning modalities, enriching both formal and informal educational environments. However, its widespread adoption faces several barriers, including high implementation costs, health and safety concerns, limited curricular integration, and insufficient educator training. Proposed solutions include strategic partnerships with technology providers and the use of cloud-based infrastructures to reduce costs and complexity. This study offers actionable insights for educators, policymakers, and developers, providing a robust, data-driven foundation for the effective and equitable integration of VR in education.

**Keywords** Virtual reality, Immersive learning, Bibliometric analysis, Educational technology, VR in education, Global trends, Challenges in education, Quality education

## 1 Introduction

Virtual reality (VR) is an advanced technology that enables users to interact directly with three-dimensional (3D) environments, providing immersive experiences through specialized hardware, such as headsets and motion sensors [144]. By generating a heightened sense of presence, VR facilitates direct interaction with digital objects within



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simulated settings. The integration of multisensory stimuli—including visual, auditory, and tactile inputs—further reinforces experiential learning. These dynamic interactions significantly enhance immersion and foster student engagement with content in authentic learning contexts [38, 52, 53]. Due to its adaptable and innovative characteristics, VR has been applied across various domains, including healthcare, engineering, education, and entertainment. It has emerged as a transformative tool with the potential to redefine learning experiences and pedagogical methodologies [75].

Although virtual reality (VR) is now widely acknowledged for its educational potential, its early development and applications were predominantly situated within the entertainment industry [140], particularly in video gaming [41]. Nevertheless, VR has demonstrated far-reaching implications that extend beyond its initial novelty, particularly within educational contexts. Unlike traditional multimedia learning methods, which rely heavily on static two-dimensional visuals, VR fosters interactive and immersive environments that can reduce cognitive load by enabling learners to engage with content dynamically and intuitively through direct experience [5]. This modality supports various learning styles—visual, auditory, and kinesthetic—offering personalized and contextually relevant educational experiences [44]. As VR becomes increasingly integrated into educational institutions, it is imperative to systematically evaluate its pedagogical effectiveness and establish evidence-based best practices to support its seamless incorporation into instructional frameworks, thereby maximizing its transformative potential in teaching and learning.

Building on these pedagogical foundations, virtual reality (VR) has been increasingly integrated into educational settings since the early 2010s, enabling innovative applications such as the simulation of scientific experiments, practical skills training in fields like medicine and engineering, and the creation of immersive environments for distance learning [57]. As hardware costs decline and accessibility improves, VR is anticipated to play a pivotal role in education by offering personalized and context-specific learning experiences tailored to diverse learner preferences and needs (P. [81, 82]). A recent report by Fortune Business Insights [48] that the VR market in education will reach USD 17.18 billion by the end of 2024, driven by widespread institutional adoption and the falling costs of VR devices. This substantial growth highlights the potential of VR to transform educational methodologies and enhance student learning outcomes globally. While this momentum reflects substantial progress, it is essential to acknowledge that the early development of virtual reality (VR) faced considerable challenges, particularly high implementation costs and complex technical requirements [97]. Advances in mobile-based systems have significantly lowered these barriers, while recent innovations—such as motion tracking and haptic feedback [152]—have further enhanced user immersion and interactivity [17]. As a result, VR now facilitates highly interactive and realistic learning experiences across a broad spectrum of disciplines, including STEM education [36, 44], chemistry [113], physics [21], biology [15, 91], mathematics [18], geography [129], language learning [66], and computer programming [137, 138].

Despite the rapid advancement of virtual reality (VR) technology and its increasing adoption in educational settings, the academic literature remains fragmented and lacks a comprehensive scope. Many existing studies are confined to specific academic domains—most notably health professions education [72]—or rely on limited data sources such as Google Scholar and PubMed. These constraints hinder the development

of a nuanced understanding of global trends and challenges associated with VR adoption in education [103]. Furthermore, the inconsistent quality control of Google Scholar raises concerns about including non-peer-reviewed publications, which may compromise the reliability and validity of literature analyses [84].

Although prior bibliometric investigations—such as those conducted by Magaña et al. [90], Rojas-Sánchez et al. [118], and Soto et al. [134]—have identified notable trends, including the exponential rise in VR-related publications and the dominance of countries such as the United States, the United Kingdom, and Chinese Taipei, these studies are often constrained by narrow scopes or regional foci. Consequently, they fail to provide a holistic understanding of VR's global trajectory and challenges across diverse educational contexts.

The present study undertakes a comprehensive bibliometric analysis that spans a broader temporal range and incorporates multiple internationally recognized databases to address these limitations. This approach facilitates a more nuanced and holistic examination of VR's educational role. Furthermore, the scarcity of systematic reviews raises concerns about selection bias in the existing literature, potentially obscuring critical insights and constraining our understanding of VR's broader impact across disciplines and educational levels [154].

Accordingly, this study analyzed publications retrieved from the Web of Science Core Collection, which comprises the Science Citation Index Expanded (SCIE), the Social Sciences Citation Index (SSCI), and the Emerging Sources Citation Index (ESCI). Only articles concurrently indexed in the Web of Science and Scopus databases were included to ensure balanced and representative coverage, although all bibliometric data were extracted exclusively from the Web of Science. This dual-indexing criterion strengthens the reliability and robustness of the dataset. Additionally, the analysis focused on publications from prominent academic publishers, including Springer Nature, Elsevier, MDPI, and IEEE, which are consistently indexed in both databases and are widely recognized for their scholarly rigor and high editorial standards.

By integrating data from these reputable sources, this study maps global trends, analyzes research patterns, and identifies key challenges and opportunities associated with implementing VR in education. The objective is to present a holistic perspective that captures diverse scholarly viewpoints and mitigates potential bias from limited database coverage, thereby supporting a more comprehensive and balanced literature review.

While previous studies have examined specific aspects of VR integration in educational settings, highlighting benefits such as increased student engagement, improved conceptual understanding, and enhanced interactivity [5, 122]. However, few studies have explored broader challenges and limitations systematically. This research aims to fill that gap by providing a comprehensive bibliometric analysis that deepens our understanding of VR's educational impact and transformative potential across diverse contexts.

A bibliometric approach provides valuable insights by examining publication trends, keyword co-occurrence, and citation patterns within the VR-in-education literature [3, 11, 121]. This study aims to uncover major research trajectories, thematic developments, and persistent barriers by addressing these gaps. To achieve this, the following research questions are proposed to guide a comprehensive exploration of the global VR research landscape and its educational implications:

**RQ1** What are the publication trends in VR-related educational research from 2010 to 2024?

**RQ2** How is VR-related research distributed across educational levels (primary, secondary, and higher education) and disciplinary domains?

**RQ3** What are the primary research themes and emerging keywords in the VR education literature?

**RQ4** What are the key challenges—technical, pedagogical, economic, and ethical—associated with implementing virtual reality (VR) in education, and what solutions have been proposed in the literature to address these challenges?

**RQ5** How do citation patterns and international collaborations shape VR's academic impact and research quality in education, and what bibliometric indicators (e.g., citation counts, journal impact factors, co-authorship networks) are used to assess these dimensions?

This study is particularly relevant for educators, policymakers, and technology developers seeking to integrate VR effectively into pedagogical practice while addressing enduring barriers to adoption. By leveraging large-scale bibliometric data and uncovering emerging research trajectories, the study provides actionable insights that inform future directions for developing, implementing, and critically evaluating VR in diverse educational contexts.

## 2 Methodology

### 2.1 Research design and data sources

This study employed a bibliometric analysis to systematically map the scholarly landscape concerning virtual reality (VR) use in education. Bibliometric analysis was chosen to assess prominent scientific publications quantitatively and uncover trends, citation structures, and knowledge gaps across disciplinary and geographic boundaries [133]. Unlike systematic reviews, which synthesize findings from a curated selection of studies to answer narrowly defined research questions, bibliometric analysis offers a broader, structural perspective by examining the evolution and configuration of an entire research domain [11].

Given this study's aim—to identify global trends and challenges in VR integration across diverse educational levels and disciplines—bibliometric analysis represents the most suitable methodological approach. It enables the evaluation of research productivity, scholarly impact, collaboration patterns, and thematic developments [42, 54], thereby contributing to a comprehensive understanding of the field.

The selection and analysis processes followed PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines to ensure a systematic and transparent approach [100, 109, 142]). The data were sourced from three primary databases in the Web of Science Core Collection: the SCIE, SSCI, and ESCI, which provide comprehensive coverage and high-quality content. The articles analyzed were published by eight reputable publishers, Springer Nature, Elsevier, MDPI, Taylor & Francis, Wiley, Frontiers, IEEE, and SAGE, selected since the rationale outlined in the introduction [68].

These publishers were selected for their strong representation of high-quality, widely recognized journals in educational technology and related disciplines.

## **2.2 Identification and data collection**

### **2.2.1 Article selection criteria**

The articles included in this study were selected based on predefined inclusion criteria to ensure the dataset's relevance, methodological rigor, and thematic coherence. The analysis spans publications from January 2010 to August 2024, capturing a critical period in the integration of virtual VR within educational contexts. The year 2010 was chosen as the starting point due to pivotal advancements in VR hardware and software that catalyzed pedagogical adoption and spurred heightened academic interest [25, 55, 63, 124]. Prior to this period, the application of VR in education was relatively marginal, characterized by isolated efforts and limited scholarly attention.

Bibliometric analysis revealed that only articles published from 2018 onward were consistently indexed in high-impact databases such as Scopus and Web of Science, indicating a growing convergence between VR research and internationally recognized academic standards. This trend reflects the field's maturation and an increasing emphasis on producing peer-reviewed research with educational significance.

This selected temporal range includes the most recent innovations, pedagogical trends, and implementation challenges, providing a comprehensive and up-to-date representation of the field's evolution. To maintain thematic consistency, only articles explicitly focused on the application of VR in educational or learning contexts were included. Eligible studies encompassed a variety of research designs, including empirical investigations, experimental studies, case reports, and literature reviews. Articles primarily concerned with technical system development or focused exclusively on entertainment-based applications were excluded as they fall outside the scope of this study's educational objectives.

### **2.2.2 Keywords and search query**

A targeted search query was constructed and applied within the selected databases to ensure the relevance and thematic alignment of the retrieved articles with the study's objectives. The search string used was: {TI = [{"virtual reality" OR VR) AND ("education" OR "learning")}] AND PY = (2010–2024). In this query, TI (Title) restricts the results to articles whose titles explicitly include the terms “virtual reality” or “VR” in conjunction with either “education” or “learning.” The PY (Publication Year) filter confines the results to publications from 2010 to 2024, capturing foundational developments and recent advances in VR applications within educational settings.

This query yielded 1220 articles that met the indexing criteria of the Science Citation Index Expanded (SCIE), Social Sciences Citation Index (SSCI), and Emerging Sources Citation Index (ESCI). These articles were published by eight leading academic publishers, selected for their reputational standing and consistent indexing in major databases. The resulting dataset constituted the basis for the subsequent bibliometric analysis.

### **2.2.3 Content analysis method**

In parallel with the bibliometric analysis, a qualitative content analysis was conducted to explore the underlying themes, topics, and challenges associated with implementing

virtual reality (VR) in education. This method involved a systematic review of the textual content from the selected articles, whereby relevant passages were coded and categorized based on emergent keywords, recurrent thematic structures, and discernible patterns.

This analysis aimed to provide deeper interpretive insight into the evolving discourse surrounding VR in educational contexts, including frequently reported challenges—such as technical limitations, pedagogical constraints, or ethical concerns—and the corresponding solutions proposed in the literature. By integrating this qualitative layer, the study aimed to complement the quantitative bibliometric findings with a context-sensitive understanding of the field's development.

#### **2.2.4 Keyword extraction process**

The keyword extraction process used RStudio, employing the Bibliometrix package and its Biblioshiny interface to systematically identify relevant terms from article titles, abstracts, and author-supplied keywords. A manual filtering procedure was implemented to review, refine, and consolidate the keyword dataset, ensuring the quality and contextual relevance of the extracted terms.

Given the high volume of generated terms, this step presented several challenges. Nevertheless, particular emphasis was placed on identifying high-frequency keywords that reflected dominant research themes across the corpus. A factorial analysis was conducted to enhance the analytical robustness of the keyword set. This technique facilitated the identification of underlying thematic structures and semantic relationships among keywords.

The factorial analysis enabled the prioritization of conceptually salient terms while eliminating those that were redundant or marginally relevant. This rigorous methodological process refined the final keyword set to accurately represent the core research topics and emerging trends within the VR-in-education literature. This curated list subsequently served as a foundational input for advanced bibliometric analyses..

### **2.3 Data screening and eligibility**

#### **2.3.1 Article screening criteria**

In the article screening process, several inclusion criteria were applied to ensure that only articles meeting specific standards were included in the analysis.

1. Criterion 1: English Language.

Only articles written in English were considered to maintain consistency in the understanding and interpretation of the data and to facilitate cross-study comparisons. Of the 1220 articles, 2 (two) were excluded because they were not written in English.

2. Criterion 2: Type of Publication.

Only original research articles were selected, excluding book chapters, reports, or conference papers. This analysis was based on substantial and in-depth scientific contributions. A total of 292 articles, including reviews or conference reports, were excluded to maintain the quality of the analysis.

3. Criterion 3: Open Access.

Articles were included only if they were open-access or had full-text availability. This criterion was adopted to ensure that all selected studies could be accessed and evaluated without restriction, promoting transparency and openness in scientific

inquiry. During the screening process, 411 articles were excluded due to a lack of full-text access.

While this exclusion aligns with the study's commitment to research accessibility, it may introduce open-access bias, wherein freely available studies are overrepresented and potentially relevant paywalled publications are underrepresented. To mitigate this limitation, several strategies were employed. First, efforts were made to ensure that the remaining full-text articles reflected diverse research on VR in education, including studies published in reputable journals by established academic publishers. This approach sought to minimize the dominance of specific open-access sources or research networks. Additionally, although the indirect inclusion of inaccessible articles—via title, abstract, or keyword-level analysis—was considered, it was ultimately rejected due to its inability to support the depth of content analysis required by this study. Although excluding inaccessible articles may have marginally limited the breadth of the sample, the final dataset remains comprehensive and methodologically robust. The selection process ensured that the included studies were of high academic quality, peer-reviewed, and representative of critical developments and challenges in the field.

### **2.3.2 Screening process**

The initial step in the article screening process involved filtering based on titles and abstracts to assess the relevance of each publication to the study's focus on VR in education. Articles that explicitly addressed the application of VR within educational contexts were retained for further evaluation, while those deemed irrelevant, not original research, or not meeting the language and accessibility criteria were excluded. These exclusions included general reports, non-peer-reviewed materials, and articles without full-text access.

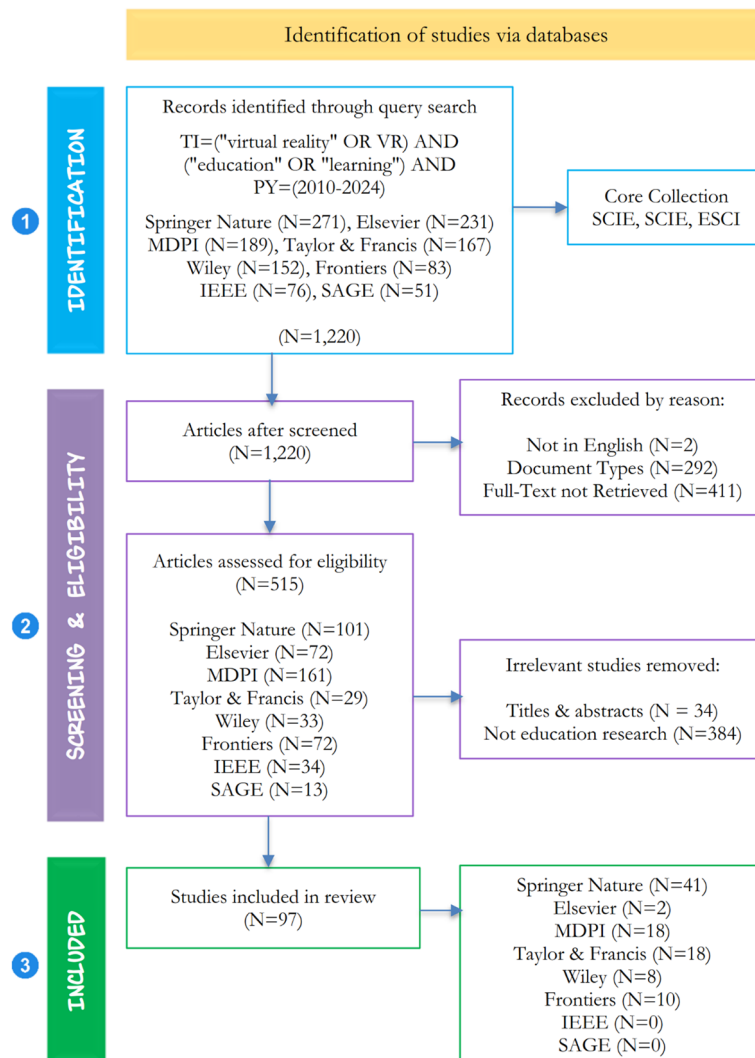
Among the 515 articles, 34 were excluded because of irrelevance to the topic of VR in education after their titles and abstracts were reviewed. Additionally, 384 articles were removed for not meeting the criteria of educational research, such as not focusing on the application of VR in an educational context. After this thorough screening, 97 articles met all the inclusion criteria and were selected for full-text evaluation.

### **2.4 Final data (Included)**

After the initial screening, the remaining 97 articles were subjected to a full evaluation stage to ensure their suitability. This evaluation thoroughly reviewed the full texts to assess the content's relevance, quality, and alignment of the content with the research focus. Articles that met all the predefined criteria—original research, open access, and a focus on VR in education—were included in the final analysis. Overall, the final screening results provide a clear overview of current research trends and the quality of the literature regarding the use of VR in education. These data were used to analyze trends, challenges, and developments in the field and provide insights for future research, as depicted in Fig. 1.

### **2.5 Data analysis**

The data analysis in this study involved several complementary approaches to provide deeper insights into the literature on virtual reality (VR) in education. Keyword analysis was conducted to identify the most frequently occurring terms, revealing dominant



**Fig. 1** PRISMA flow diagram

themes and commonly explored research areas. Citation analysis examined citation patterns and research impact, providing insights into the influence and scholarly reach of individual studies. Productivity analysis evaluated the contributions of individual authors and institutions, highlighting the most prolific contributors to VR-related educational research. Trend analysis was employed to examine temporal changes in the literature, facilitating the identification of shifts in research focus and the evolving application of VR technology in educational contexts. Finally, network visualization was performed using RStudio software to map relationships and collaboration patterns among authors, institutions, and countries. This visual mapping illustrated the structural connections within the research landscape and revealed key nodes of scholarly influence.

## 2.6 Methodological tools and software

Several software tools were effectively utilized to analyze the data. Microsoft Excel was used for initial data management and fundamental statistical analysis, facilitating the creation of tables, graphs, and documentation. RStudio, with the Bibliometrix package (<https://www.bibliometrix.org/>), is critical in conducting in-depth bibliometric analysis,

including keyword analysis, citation analysis, and thematic evolution [11]. Python was employed to perform supplementary data analysis and to generate custom visualizations using libraries such as Pandas and Matplotlib. Microsoft Power BI was used to create dynamic visualizations of keyword distributions and temporal trends, thereby enhancing the interpretability of the findings for diverse audiences. The integration of these tools ensured a comprehensive, accurate, and visually compelling analytical process, contributing to the depth and clarity of the results.

2.7 Quality and credibility assurance

To ensure the validity and reliability of the analysis, data collection was conducted meticulously, and only articles meeting the predefined quality criteria were included. The completeness of bibliographic metadata was also verified to confirm the accuracy and consistency of key elements, including article titles, author names, publication years, keywords, and citation counts. This verification step was essential for maintaining the integrity of the dataset used in the bibliometric analysis.

Table 1 presents the analysis of the completeness of the bibliographic metadata for the articles used in this bibliometric study. The data show that most metadata elements, such as title (TI), author (AU), publication year (PY), and total citations (TCs), are fully complete, with a 0% deficiency across all these categories. These findings indicate that the data possess high integrity and strong reliability for bibliometric analysis.

However, there are minor deficiencies in the "Keywords" (DE) and "Keywords Plus" (ID) categories. The "Keywords" category has a deficiency of 2.06%, whereas "Keywords Plus" has a deficiency of 6.19%. Although this does not significantly affect the overall analysis, it does suggest that some articles did not include essential keywords that would provide a deeper understanding of the research topics.

Notably, there is a complete lack of data in the "Science Categories" (WC) category, where 100% of the data are unavailable. The absence of this information could limit the analysis of literature classifications and the detailed distribution of scientific disciplines. To address this issue, we conducted an in-depth review of all the articles and manually classified them into scientific categories based on their subject matter. This step allows

**Table 1** Completeness of Bibliographic Data

| Metadata | Description          | Missing Counts | Missing % | Status             |
|----------|----------------------|----------------|-----------|--------------------|
| AB       | Abstract             | 0              | 0.00      | Excellent          |
| C1       | Affiliation          | 0              | 0.00      | Excellent          |
| AU       | Author               | 0              | 0.00      | Excellent          |
| CR       | Cited References     | 0              | 0.00      | Excellent          |
| RP       | Corresponding Author | 0              | 0.00      | Excellent          |
| DI       | DOI                  | 0              | 0.00      | Excellent          |
| DT       | Document Type        | 0              | 0.00      | Excellent          |
| SO       | Journal              | 0              | 0.00      | Excellent          |
| LA       | Language             | 0              | 0.00      | Excellent          |
| PY       | Publication Year     | 0              | 0.00      | Excellent          |
| TI       | Title                | 0              | 0.00      | Excellent          |
| TC       | Total Citation       | 0              | 0.00      | Excellent          |
| DE       | Keywords             | 2              | 2.06      | Good               |
| ID       | Keywords Plus        | 6              | 6.19      | Good               |
| WC       | Science Categories   | 97             | 100.00    | Completely missing |

us to better understand the distribution of disciplines despite the absence of original metadata.

While there are some deficiencies in specific metadata categories, the available data are sufficiently complete to conduct a comprehensive bibliometric analysis. Additional steps are taken to address the existing gaps and ensure the accuracy of the analytical results.

### 3 Results

#### 3.1 Main information

After completing the screening and evaluation process, the initial search, covering January 2010 to August 2024, was refined to focus on articles published between 2018 and 2024. This adjustment ensured that the analysis captured the most relevant and recent trends in VR research within education.

The primary reason for narrowing the timeframe was that articles published before 2018 were found to be less aligned with the current standards and focus of VR research in education. As VR technology and its applications in education have evolved significantly in recent years, earlier studies did not fully reflect these advancements and emerging trends.

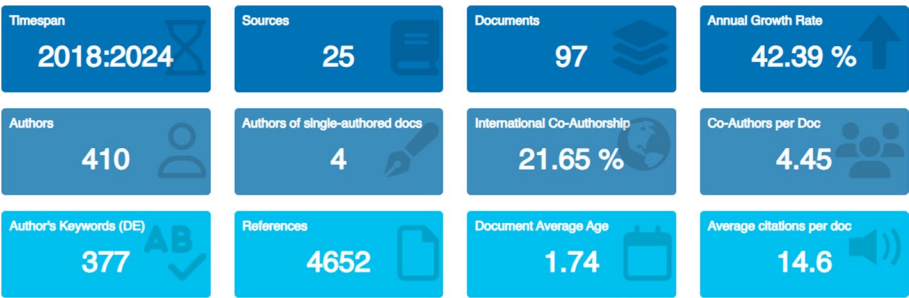
Furthermore, data analysis revealed a sharp increase in VR research activity in education starting around 2018, marked by an annual growth rate of 42.39%. This surge highlights the growing integration of VR in educational contexts. By concentrating on the more recent period, the study was able to capture the latest developments better and ensure that the findings are grounded in current trends and innovations.

During the selection process, articles that did not meet specific relevance criteria or were not directly related to VR in educational settings were excluded. This rigorous screening process ensured that the analyzed studies were both pertinent and of high quality, maintaining the integrity of the research. As a result, many earlier articles which lacked relevance to the evolving research focus were excluded.

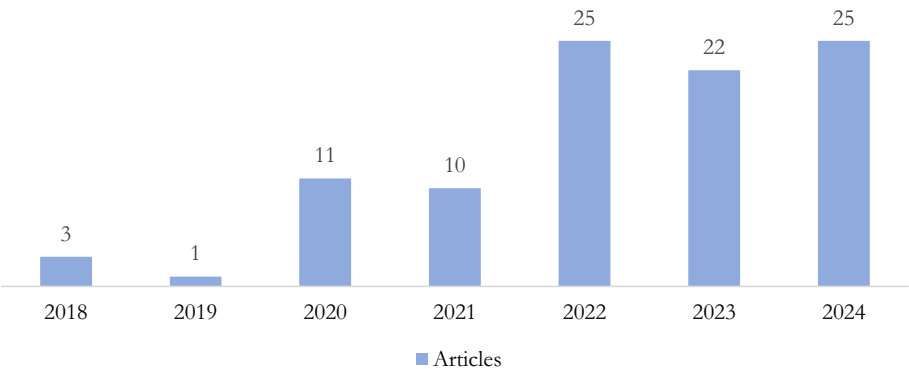
By limiting the scope to articles published between 2018 and 2024, the study offered a more precise representation of VR's current role in education, highlighting the most recent advances and challenges in the field. This approach provided a clear picture of the direction of VR research and the contributions made in the last few years.

The final results of this process included 97 articles that met all the established criteria. These articles were selected based on their relevance to the topic of VR in education, the quality of the research, and the accessibility and language of publication. As mentioned, these articles were published in 25 journals from eight reputable publishers. The average document age was 1.74 years, indicating that most of the articles were recent publications providing a current perspective on the development of VR in education. This is illustrated in Fig. 2, which provides an overview of the main information regarding the final data.

The bibliometric data reflect the field's diversity and collaborative nature. The 97 articles collectively include 226 "Keywords Plus" and 377 author-assigned keywords, highlighting the multidimensional scope of VR research in education. Regarding authorship, 410 researchers contributed to the selected publications, with an average of 4.45 co-authors per document, demonstrating a high level of collaborative scholarship.



**Fig. 2** Main information of the final data



**Fig. 3** Overview of publication trends

Moreover, the international collaboration rate is 21.65%, indicating a significant level of global engagement in this research domain.

3.2 Trends in publications (RQ1)

There was a significant increase in publications from 2018 to 2024. In 2018, there were only three publications, whereas in 2022 and 2024, the number of publications surged to 25. This reflects the growing interest and substantial growth in research on VR in education. Fig. 3 presents an overview of publication trends related to the use of VR in education from 2018 to August 2024.

Notable year-to-year increases occurred in 2019, 2020, and 2021, with the number of publications rising from one in 2019 to eleven in 2020 and ten in 2021. These fluctuations may reflect the field’s early development and growing adoption of VR technology in response to global shifts—notably the COVID-19 pandemic—which accelerated the integration of digital tools in education [34]. During this time, many institutions sought alternative modes of teaching and learning, potentially stimulating increased exploration of immersive technologies such as VR.

The publication output remained relatively stable in 2022 and 2023, with minor fluctuations, and continued at a comparable level into 2024 (as of August). This sustained output level suggests that scholarly engagement with VR in education has maintained momentum beyond the pandemic-driven surge.

Overall, the publication trend demonstrates the field’s dynamic evolution, characterized by an initial period of rapid growth—partly influenced by external disruptions—followed by a phase of consolidation. The brief decline in 2019 may reflect early uncertainty or transitional challenges in integrating VR into educational research and practice.

### 3.2.1 Distribution by multiple indices

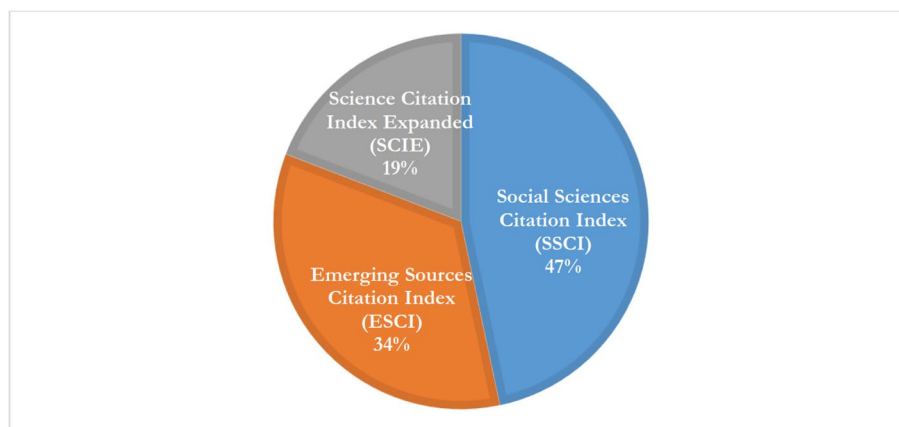
Figure 4 illustrates that most articles that met the selection criteria were published in the Social Sciences Citation Index (SSCI), with 56 articles highlighting the dominance of the social sciences in VR research within education. This finding indicates that VR in education, including human interaction, education, and technology studies, is highly relevant and widely accepted in the social sciences community. The prominence of the SSCI suggests that the research is likely focused on the pedagogical, psychological, and social aspects of VR application in education.

A total of 41 articles were published in journals indexed by the Emerging Sources Citation Index (ESCI), reflecting increasing scholarly interest in the application of virtual reality (VR) in education within emerging or developing academic outlets. ESCI comprises journals under evaluation for inclusion in more established citation indices and frequently represent innovative or evolving research domains. The relatively high proportion of ESCI-indexed articles suggests that VR in education, although still an emerging area of inquiry, is rapidly gaining scholarly recognition and is being actively investigated in journals positioned at the forefront of academic innovation and disciplinary expansion.

Meanwhile, 23 articles appeared in journals listed in the Science Citation Index Expanded (SCIE), which covers high-impact journals in science and technology. Although this number is lower than those in SSCI and ESCI, it highlights the field's relevance to the technical and scientific domains. The presence of SCIE-indexed articles highlights the significance of VR's technological foundations and its impact on innovation in educational tools, systems, and interfaces. Together, these findings illustrate the interdisciplinary nature of VR in education, encompassing both pedagogical inquiry and technological advancement.

It is important to note that a single article can be indexed in multiple bibliometric databases. In this study, the distribution of articles across the SSCI, SCIE, and ESCI reflects such overlaps, as many journals are listed in more than one index. Consequently, the same article may be counted multiple times in index-specific statistics, which should be interpreted accordingly.

All 97 articles included in this study were also indexed in Scopus, one of the significant global bibliometric databases. This cross-indexing serves as an additional validation layer, confirming that the dataset is comprehensive and representative of research



**Fig. 4** Distribution of articles by index

acknowledged across leading academic platforms. Consequently, including Scopus-indexed articles enhances the credibility, robustness, and international relevance of the research.

The inclusion of SSCI, SCIE, ESCI, and Scopus in this bibliometric study offers several important advantages:

- (1) Each index has a distinct focus and coverage. SSCI specializes in the social sciences literature, SCIE encompasses a wide range of scientific disciplines, and ESCI includes emerging sources that may not yet be listed in the major indices. By utilizing all three indices, this study can provide a more comprehensive view of the distribution and impact of articles across multiple disciplines and sources.
- (2) Each index maintains rigorous selection criteria for including journals and articles. Using multiple indices ensures that the articles analyzed come from journals verified for quality across various fields, increasing the reliability and credibility of bibliometric analysis.

This multi-index approach facilitates a more robust and comprehensive bibliometric analysis. It offers a nuanced perspective on the research landscape of VR in education, capturing its interdisciplinary nature and ensuring that findings are grounded in a high-quality, internationally recognized body of literature.

3.2.2 Distribution by top-eight publishers

Figure 5 shows Springer Nature is the leading publisher, contributing 41 articles, representing 42.27% of the total publications. This substantial proportion underscores Springer Nature's pivotal role in advancing and disseminating research on VR in education. The high volume of articles published by Springer Nature highlights its significant influence in shaping the discourse and driving progress in this field.

The distribution of articles across publishers revealed that MDPI and Taylor & Francis each contributed 18 articles, accounting for 18.56% of the total publications. Frontiers Media published ten articles, representing 10.31% of the dataset, whereas Wiley contributed eight articles, or 8.25%. Elsevier had a notably lower contribution, with only two articles (2.06%), and neither IEEE nor SAGE provided any articles for this analysis.

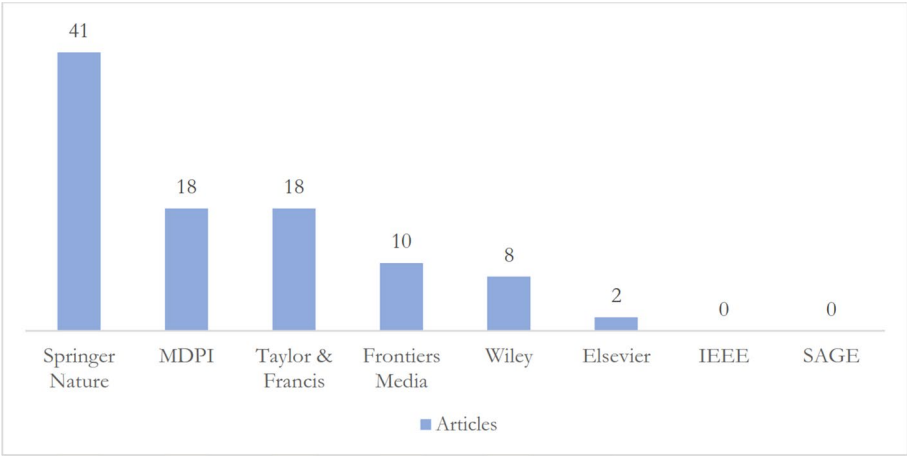


Fig. 5 Distribution of articles by leading publisher

Importantly, this distribution pertains specifically to the 97 articles that met the study's criteria and does not assess the overall contribution of each publisher. This analysis is based on articles indexed in multiple recognized databases, including the Web of Science and Scopus, while articles indexed in only one database were excluded. Consequently, the contributions of other publishers may not be fully represented in this dataset despite their potential relevance to the field. However, the focus on articles indexed in multiple well-regarded databases ensures the consistency and credibility of the data used in this study.

### 3.2.3 Regional and institutional contributions

Table 2 provides a detailed overview of the geographic distribution of publications, offering key insights into the global contributions to VR research in education. The United States has emerged as the foremost contributor, with ten articles highlighting its preeminence in this field. This leadership is mainly attributable to the country's robust research infrastructure, substantial funding, and esteemed academic institutions dedicated to technology and innovation. Prominent universities such as MIT, Stanford, and the University of California are at the cutting edge of educational technology and VR research, generating a significant volume of high-quality studies. A key driver of the United States' dominance is its strong culture of academia-industry collaboration. Leading technology companies, including Google, Meta (formerly Facebook), and Microsoft, frequently partner with universities to explore applied research in immersive learning. Such partnerships foster practical innovation and are associated with high-impact outcomes, as evidenced by Evans et al. [45]. Bhullar et al. [19] further note that these collaborations enhance research productivity more significantly than teaching effectiveness, underscoring their critical role in the digital transformation of higher education.

Germany, the Netherlands, and China each contributed nine articles, indicating that these countries also play a significant role in the field. Germany is renowned for its robust infrastructure in technological research and innovation. The country has several prestigious academic institutions and research centers highly regarded in technology. For example, the Technical University of Munich and the Fraunhofer Society are two institutions actively engaged in cutting-edge VR research. Additionally, Germany integrates VR into its education and professional training systems, as demonstrated in projects such as "Virtual Reality for Learning and Teaching," which focuses on developing and applying VR in educational contexts (<https://www.itwm.fraunhofer.de/en/departments/mf/technikum/virtual-reality-labor.html>).

**Table 2** Distribution by top ten countries

| No | Countries      | Articles |
|----|----------------|----------|
| 1  | USA            | 10       |
| 2  | GERMANY        | 9        |
| 3  | NETHERLANDS    | 9        |
| 4  | CHINA          | 9        |
| 5  | UNITED KINGDOM | 7        |
| 6  | AUSTRALIA      | 6        |
| 7  | FINLAND        | 6        |
| 8  | SOUTH KOREA    | 6        |
| 9  | TAIWAN         | 6        |
| 10 | GREECE         | 5        |

The Netherlands is also a key hub for technology research, with several leading universities and research institutions actively involved in VR. Institutions such as Delft University of Technology, Eindhoven University of Technology, and others in the Netherlands are engaged in research projects that leverage VR for various applications, including education. The Netherlands is known for its innovative education and research approach and strong collaboration between academia and the tech industry. The data indicate that the Netherlands has various initiatives, such as “virtual reality in education,” which integrates technology with learning methodologies. This collaborative environment has fostered advancements in VR research and its educational applications, making the Netherlands a prominent player in this domain, as reflected in Table 2 shows the distribution of the top 10 countries.

With an equal contribution to the number of publications, China has experienced rapid growth in its VR technology and research adoption. Several key factors drive this growth:

- (1) The Chinese government has prioritized VR as a strategic component of national technology development plans. The “Made in China 2025” initiative and the “13th 5 Year Plan” highlight VR as essential for industrial and educational transformation, emphasizing technological innovation and development.
- (2) Leading Chinese universities, such as Tsinghua University and Peking University, are heavily involved in significant VR research. These studies span a range of applications, including educational simulations and the development of VR-based learning tools.
- (3) China’s expanding technology ecosystem, with major tech companies such as Huawei and Tencent investing in VR, supports rapid advancement. These companies are developing VR hardware and software and collaborating on research with academic institutions.
- (4) Given China’s vast population and the urgent need to modernize its education system, VR technology is being embraced to enhance educational quality. This drive creates numerous research opportunities and practical applications, leading to substantial publications.

With seven articles, the United Kingdom (UK) reflects its active involvement in VR research within education. The UK is home to leading universities such as University College London and the University of Cambridge, both at the forefront of developing and applying VR technology in educational contexts. The UK also benefits from policies that support innovative research and international collaboration, such as the Horizon Europe program, which funds research projects in educational technology.

Australia, Finland, South Korea, and Taiwan contributed six articles and demonstrated strong commitment in this field. Australia is recognized for its innovative approach to education and technology, with institutions such as the University of Sydney and the University of Melbourne actively involved in VR research. Finland, known for its progressive education system, has integrated VR technology into its educational strategy, with universities like the University of Helsinki leading research in this area. Along with leading tech companies, South Korea, supported by government policies in technology and education, plays a significant role in the development and application of VR. With its growing academic institutions and technology industry, Taiwan has also contributed

to VR research, reflecting its efforts to advance educational technology (as illustrated in Fig. 6).

Although Greece contributed five articles, it still showed significant engagement. The country has participated in VR and educational technology research through institutions such as the National Technical University of Athens, even though its contribution to this area is smaller than that of other countries on this list [22].

### 3.2.4 Corresponding author countries

China leads with 14 articles, underscoring its significant contribution to educational VR research. Germany, South Korea, the Netherlands, and the United Kingdom follow, each contributing between 6 and 8 articles. The USA has five articles, while Australia, Iran, Spain, and Finland have also made notable contributions. Notably, China has the highest number of corresponding domestic authors, with 13 out of 14 articles indicating that most related research was conducted within the country. In contrast, Finland exhibits a higher level of international collaboration, with 2 out of 3 articles involving corresponding international authors.

### 3.2.5 Countries with the most citations

The United Kingdom has the highest total number of citations (281) and a notably high average citation rate per article (46.80), indicating that its research is highly valued and has significantly impacted the global academic community. In contrast, despite its substantial volume of publications, China has a lower total citation count (139) and a much lower average number of citations per article (9.90). This disparity suggests that while China is prolific in its output, its research may not have achieved the same global influence as countries with higher average citation rates. Spain also has a strong citation profile with a high total citation count (129) and a robust average citation rate (32.20), indicating the quality and impact of its research. Additionally, countries such as Denmark and New Zealand have very high average citation rates, reflecting the significant influence of their research despite their smaller volume of publications.



**Fig. 6** Distribution of articles by country

This analysis highlights the differences in the impact and international recognition of research conducted across various countries. For instance, the United Kingdom and Denmark demonstrate a high impact on citations per article, while China has a lower average citation rate despite its large volume of publications. This reflects the complexity of assessing research quality and impact across different nations, emphasizing the importance of considering the quantity of publications and their influence within the scientific community.

3.2.6 Distribution by journals

Table 3 provides an overview of the distribution of articles based on the journals that published them, offering insights into which journals are most active in disseminating research related to VR in education. *BMC Medical Education* led with 20 articles (20.62% of the total), indicating a strong interest in integrating VR within medical education. The journal also has a relatively high total citation count (TC) of 199, reflecting its significant influence on the academic community. *Education Sciences* ranks second with 18 articles (18.56%) and a total citation count of 131, confirming its role as a leading journal in educational technology. *Frontiers in Education* follow with ten articles (10.31%) but have a lower total citation count (49), possibly indicating a more significant niche or emerging focus on VR-related publications. *Interactive Learning Environments* and the *International Journal of Educational Technology in Higher Education* contributed six articles (6.19%), with a substantial total of 119 and 104 citations, respectively, showing that VR research in interactive learning and higher education also garners significant attention. The *Smart Learning Environments* and *Education and Information*

**Table 3** Distribution by Journals

| No | Journals                                                            | Docs | TC  |
|----|---------------------------------------------------------------------|------|-----|
| 1  | BMC Medical Education                                               | 20   | 199 |
| 2  | Education Sciences                                                  | 18   | 131 |
| 3  | Frontiers in Education                                              | 10   | 49  |
| 4  | Interactive Learning Environments                                   | 6    | 119 |
| 5  | International Journal of Educational Technology in Higher Education | 6    | 104 |
| 6  | Journal of Computer-Assisted Learning                               | 5    | 50  |
| 7  | Education and Information Technologies                              | 4    | 88  |
| 8  | Smart Learning Environments                                         | 4    | 159 |
| 9  | British Journal of Educational Technology                           | 3    | 31  |
| 10 | Cogent Education                                                    | 3    | 9   |
| 11 | Journal of Science Education and Technology                         | 3    | 19  |
| 12 | Educational Technology Research and Development                     | 2    | 36  |
| 13 | Educational Media International                                     | 1    | 1   |
| 14 | Educational Research Review                                         | 1    | 82  |
| 15 | European Journal of Engineering Education                           | 1    | 7   |
| 16 | Higher Education Pedagogies                                         | 1    | 25  |
| 17 | Innovations in Education and Teaching International                 | 1    | 8   |
| 18 | International Journal of Science Education, Part B                  | 1    | 3   |
| 19 | The Internet and Higher Education                                   | 1    | 59  |
| 20 | Irish Educational Studies                                           | 1    | 4   |
| 21 | Journal of Computing in Higher Education                            | 1    | 6   |
| 22 | Journal of Research on Technology In Education                      | 1    | 4   |
| 23 | Medical Education Online                                            | 1    | 9   |
| 24 | Research in Learning Technology                                     | 1    | 210 |
| 25 | Technology Knowledge and Learning                                   | 1    | 4   |

Technology contributed four articles each. However, *Smart Learning Environments* has a higher total citation count (159) than *Education and Information Technologies* (88), indicating a more substantial academic influence within its domain.

The *British Journal of Educational Technology* and several other journals, such as *Cogent Education* and the *Journal of Science Education and Technology*, contributed fewer articles but still play a key role in disseminating research related to VR in education.

Notably, some journals with only a single article, such as *Research in Learning Technology* (with a total citation count of 210) and *The Internet and Higher Education* (with a total citation count of 59), have a significant impact despite their smaller contribution in terms of document count. This indicates that while these journals may have fewer articles related to VR, those articles tend to have a very high impact, as reflected in their citation numbers.

### 3.2.7 Most relevant authors

Table 4, "Most Relevant Authors," highlights the leading contributors to VR research in education, ranked by the number of documents they have produced and the total citations (TCs) received. Meysam Siyah Mansoori holds the top position, with three documents and 199 citations. This figure suggests that Meysam Siyah Mansoori is not only prolific in terms of publication volume but also that these publications have significantly impacted the field. A high citation count may indicate the quality of the research or its strong relevance to current issues in VR research.

Fernando Russo Abegao, with two documents and 131 citations, illustrates how impactful a few high-quality publications can be in VR research. Similarly, Seyyed Mohsen Azizi and Jeong-Won Han have achieved significant influence, with 119 and 159 total citations from just two documents each. This highlights that despite fewer publications, specific authors produce highly relevant and valued work from the academic community.

In contrast, authors such as Georgios Lampropoulos, with two documents but only nine citations, may have newer publications or work that has not yet reached a broader audience. Overall, the analysis shows that while the quantity of publications is important, the quality and impact of the work, as measured by citations, are critical in shaping researchers' influence in the VR field in education.

Overall, this analysis emphasizes the importance of quantity and quality in academic contributions. High-impact authors not only produce a significant number of

**Table 4** Distribution by Most Relevant Authors

| No | Authors                  | Docs | TC  |
|----|--------------------------|------|-----|
| 1  | Meysam Siyah Mansoori    | 3    | 199 |
| 2  | Fernando Russo Abegao    | 2    | 131 |
| 3  | Friday Joseph Agbo       | 2    | 49  |
| 4  | Seyyed Mohsen Azizi      | 2    | 119 |
| 5  | Athanasios Christopoulos | 2    | 104 |
| 6  | Pamela Cowan             | 2    | 50  |
| 7  | Rachel Farrell           | 2    | 88  |
| 8  | Jeong-Won Han            | 2    | 159 |
| 9  | Mikko-Jussi Laakso       | 2    | 31  |
| 10 | Georgios Lampropoulos    | 2    | 9   |

publications but also contribute meaningfully to the advancement of knowledge within their discipline.

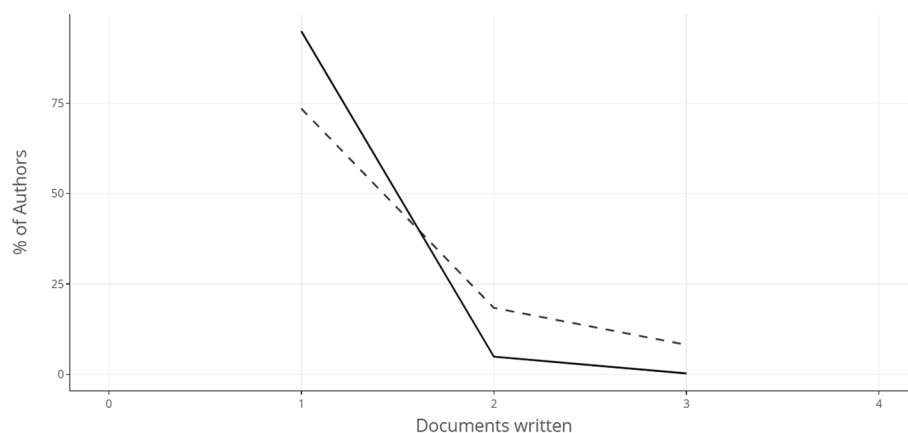
Most of the authors, 389 individuals (94.9%), had only one publication, indicating that many contributors were sporadically involved in this field. In contrast, only 20 authors (4.9%) had two publications, and just one author (0.2%) had three publications. This pattern reflects that a small group of highly productive authors contributes significantly more to publications than the majority. The distribution aligns with Lotka's Law (as shown in Fig. 7), which posits that the number of authors with  $n$  publications decreases exponentially as  $n$  increases. Therefore, while many individuals participate in VR research, only a few individuals consistently produce multiple publications, highlighting the presence of key authors who make more substantial contributions to the field.

### 3.3 Distribution across disciplines and educational levels (RQ2)

The analysis of the studies included in this review highlights the wide-ranging applications and impact of VR in education across various disciplines, including design education, chemistry, biochemistry, medicine, dentistry, biotechnology, and engineering [15, 70, 93]. This diversity underscores VR's broad potential to transform teaching methods in multiple fields. However, the attention that VR devices receive is not evenly distributed. Several fields, such as medicine and biotechnology, have garnered increased research interest, particularly in medical training [80], laboratory simulation [8], and technical skill development.

Research by Fromm et al. [47] and Speck et al. [135] illustrates how VR effectively supports practical training and the development of clinical skills. Further studies, such as those by Chen et al. [28] and Ojala et al. [107], have confirmed the effectiveness of VR in specialized training contexts, including anatomy (via VR skull models) and medical procedures, making it an invaluable tool in health education [37]. VR enables students to interact with dynamic 3D anatomical images, zoom, rotate, or even virtually deconstruct layers to explore underlying structures, enhancing their understanding of internal anatomy [111].

VR is widely used to enhance learning outcomes across education levels, from elementary to higher education. In elementary education, VR introduces foundational concepts



**Fig. 7** Lotka's Law distribution: percentage of authors vs. documents written this graph illustrates the typical pattern of authorship distribution where a small percentage of authors write many documents, while most authors write only a few documents

such as science, mathematics, and language. For example, students can virtually explore natural ecosystems or participate in virtual chemistry labs to grasp basic scientific concepts [58]. In science education, VR provides clear spatial and visual representations of otherwise abstract concepts [65], enabling more direct and compelling learning experiences.

VR allows students to experience real-world or historical situations virtually. For example, students can "visit" historical sites or explore different geographical environments, giving them a better context for understanding their study topics [14]. It can also include virtual experiences such as space travel or underwater exploration, which are physically challenging to achieve [65].

In higher education, VR is applied in more advanced contexts, such as medical and biotechnology training [98]. Marks & Thomas [95] and K. G. Kim et al. [67] emphasized how VR enhances laboratory education and professional training. In the medical field, VR offers surgical simulations and emergency scenarios [50, 140], allowing medical students to practice specific skills in a controlled environment. Surgical VR simulations, for example, provide intensive training without risking patient safety, enabling repeated practice to perfect skills [49].

Similarly, in biotechnology, VR facilitates laboratory simulations and the development of analytical techniques, helping students understand complex processes and gain practical skills. These simulations are designed to prepare students for future professional roles by offering immersive and realistic environments [16].

VR has also proven effective at enhancing student engagement and motivation through its interactive design and immersive learning experience, which traditional methods often lack. Studies by [6] and Shen et al. [129] demonstrate how VR enriches learning across various disciplines, from language to the arts. In language education, VR can create authentic social interactions in immersive environments [104]. While fostering student autonomy, reducing learning anxiety, and encouraging creativity [89], the controlled VR environment minimizes distractions that might be present in typical learning settings, helping students stay focused on tasks and concepts [58]. Overall, VR has the potential to revolutionize education by offering engaging, effective, and innovative learning experiences across various levels and disciplines.

Despite its benefits, integrating VR into the curriculum poses challenges. Sprenger & Schwaninger [136] and [29] highlighted that while VR offers many advantages, barriers such as technology acceptance and integration into existing curricula persist. Section 4: Challenges and Solutions (RQ4) will further explore these challenges. A detailed summary of the 97 reviewed studies, covering applications, research areas, and impacts, is presented in Appendix A: Summary of Research Papers Included in the Analysis.

### **3.3.1 Distribution by academic discipline**

The analysis revealed significant differences in the frequency of VR research across various disciplines. Disciplines such as medical and health education and STEM are highly active areas for VR applications, particularly in simulation-based learning. For example, Gao et al. [50] focused on using VR for surgical procedure simulations, highlighting how VR enables surgical technique training with minimal risk. Medical students can practice in highly realistic environments, reducing errors during actual procedures and

improving their practical skills [94]. Similarly, Lee et al. [79] emphasized how VR helps nursing students master clinical skills within a controlled and safe environment.

In STEM education, VR is utilized extensively in biology, chemistry, and engineering education to simulate complex experiments and technical processes. VR allows students to conduct virtual chemistry or physics experiments and safely practice engineering skills through realistic simulations [24]. Chen et al. [28] reported that VR can create immersive biological simulations, enabling students to observe and interact with biological processes such as cellular functions, metabolism, and organism interactions. Game-based VR technology also offers interactive, immersive experiences that aid in understanding human anatomy and physiology [43]. In virtual environments, users can view and interact with 3D models of organs and body systems, enhancing their understanding of human structure and function. These studies reported improved conceptual understanding and practical skills in biology. Similarly, Hu-Au & Okita [58] and [12] noted that VR enables students to visualize chemical reactions and grasp complex chemistry concepts without safety risks.

Atta et al. [12] and Kersting et al. [65] also explored VR applications in astronomy education, where VR allows students to explore solar systems, galaxies, and other astronomical objects in virtual settings. VR provides opportunities for students to interact with virtual robotic models, test designs, and robotic algorithms. Table 5 illustrates the distribution of VR research across various academic disciplines, highlighting its diverse applications in education.

Research by [75] has examined how gamification elements are applied in VR environments to enhance engagement in education. Gamified elements such as point scoring, levels, and rewards are more immersive in VR because users interact with 3D environments and virtual worlds. Unlike non-VR settings, VR provides a heightened sense of presence, which can deepen user engagement and make game-like challenges more compelling [87]. This immersive quality could benefit individuals who require practical skill development, such as engineering or medicine. For example, completing levels in a VR surgical simulation may feel more authentic and rewarding due to the lifelike environment compared with merely using a gamified 2D platform. Therefore, VR-based gamification can offer more impactful learning experiences, which may lead to greater motivation and knowledge retention among students [2, 140].

Beyond motivation, VR has been shown to support more efficient cognitive processing. While traditional multimedia tools (e.g., videos or PowerPoint slides) are limited to static or linear content, VR offers a multisensory, interactive environment. According to Allcoat & von Mühlenen [5], this immersive design can reduce cognitive load by allowing learners to navigate complex information actively rather than passively receive it. For example, students learning anatomy can manipulate 3D anatomical structures in real-time, aiding their spatial reasoning and conceptual understanding [4]. Furthermore, VR tends to reduce external distractions, enabling students to focus more deeply as they engage with the material in an environment that feels experientially authentic.

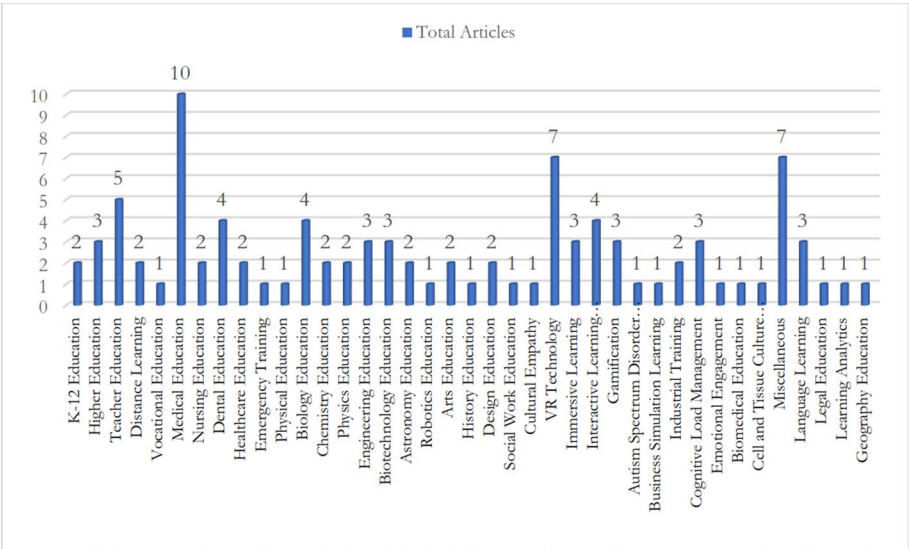
Importantly, VR has also shown promise in supporting neurodiverse learners, particularly those with autism spectrum disorder (ASD). Research by Lu et al. [88] demonstrated how VR can provide safe, structured, and repeatable learning environments, which can lower anxiety levels and enhance social communication skills in learners with ASD. However, the practical implementation of such VR solutions presents several

**Table 5** Distribution by academic discipline

| No | Academic discipline                 | Sub category                       | Articles                                  |
|----|-------------------------------------|------------------------------------|-------------------------------------------|
| 1  | Education                           | K-12 Education                     | [20, 144]                                 |
|    |                                     | Higher Education                   | [9, 47, 95]                               |
|    |                                     | Teacher Education                  | [23, 35, 46, 111, 117]                    |
|    |                                     | Distance Learning                  | [87, 149]                                 |
|    |                                     | Vocational Education               | [67]                                      |
| 2  | Medical and Health Education        | Medical Education                  | [10, 17, 49, 50, 64, 80, 93, 94, 98, 148] |
|    |                                     | Nursing Education                  | [78, 79]                                  |
|    |                                     | Dental Education                   | [31, 37, 59, 70]                          |
|    |                                     | Healthcare Education               | [107, 153]                                |
|    |                                     | Emergency Training                 | [135]                                     |
| 3  | STEM                                | Physical Education                 | [21]                                      |
|    |                                     | Biology Education                  | [43, 44, 110]                             |
|    |                                     | Chemistry Education                | [58, 113]                                 |
|    |                                     | Physics Education                  | [24, 73]                                  |
|    |                                     | Engineering Education              | [51, 137, 141]                            |
|    |                                     | Biotechnology Education            | [15, 16, 122]                             |
|    |                                     | Astronomy Education                | [12, 65]                                  |
|    |                                     | Robotics Education                 | [8]                                       |
| 4  | Arts and Humanities                 | Arts Education                     | [52],P. [81, 82]                          |
|    |                                     | History Education                  | [34]                                      |
|    |                                     | Design Education                   | [7, 53]                                   |
| 5  | Social Sciences                     | Social Work Education              | [115]                                     |
|    |                                     | Cultural Empathy                   | [14]                                      |
| 6  | Technology and Interactive Learning | VR Technology                      | [41, 83, 86, 92, 102, 132, 150]           |
|    |                                     | Immersive Learning                 | [38, 62, 146]                             |
|    |                                     | Interactive Learning Environments  | [6, 96, 105, 125]                         |
|    |                                     | Gamification                       | [2, 75, 140]                              |
| 7  | Special Education                   | Autism Spectrum Disorder Education | [88]                                      |
| 8  | Business and Vocational Training    | Business Simulation Learning       | [89]                                      |
|    |                                     | Industrial Training                | [1, 128]                                  |
| 9  | Cognitive and Behavioral Sciences   | Cognitive Load Management          | [4, 5, 39]                                |
|    |                                     | Emotional Engagement               | [71]                                      |
| 10 | Healthcare and Biomedical Sciences  | Biomedical Education               | [85]                                      |
|    |                                     | Cell and Tissue Culture Training   | [152]                                     |
| 11 | Miscellaneous                       | Miscellaneous                      | [26, 27, 40, 74, 77, 136, 147]            |
|    |                                     | Language Learning                  | [(29, 66, 104)]                           |
|    |                                     | Legal Education                    | [97]                                      |
|    |                                     | Learning Analytics                 | [33]                                      |
|    |                                     | Geography Education                | [129]                                     |

challenges. High hardware and software costs may limit access for schools with constrained budgets, and developing tailored content requires collaboration among educators, software developers, and mental health professionals to ensure educational efficacy and psychological safety.

Figure 8 visually represents the disciplinary distribution of VR research in education, highlighting dominant fields such as medical education, STEM, and social sciences, while revealing underrepresented domains, including language learning and specific areas within the humanities.



**Fig. 8** Distribution of articles by academic discipline

Future research should further explore the educational value of VR applications for language learning, which remains underexplored, and its potential in other disciplines that have not yet been fully investigated. Furthermore, the effectiveness of VR may not be consistent across different student demographics. For example, younger students may adapt more quickly to VR technology, whereas older students or those with limited technological experience may struggle with the learning curve. Additionally, there is little consensus on the long-term effects of VR on learning outcomes—most studies focus on short-term improvements in engagement or understanding. More longitudinal research is needed to assess whether VR improves knowledge retention and academic performance.

**3.4 Topics and keywords evolution (RQ3)**

Table 6 analyzes the most frequent words found in related research related to the application of VR in education, reflecting the field's core themes and focus areas. The keyword "students" indicates that much of the related research centers on how VR impacts learners in individual and collaborative educational settings. High-ranking keywords such as "technology" and "education" reaffirm that the intersection of technological advancement and its application in education is a primary focus of these studies.

Simulation is a key theme reflecting VR's widespread use in recreating real-world scenarios for skill development or deeper conceptual understanding. This theme is particularly relevant in disciplines such as STEM and medical education, where the simulation of procedures and experiments is a standard pedagogical tool [110]. As shown in Table 6, the frequency of these terms underscores their central role in current research trends, highlighting VR's potential to revolutionize hands-on learning and practical training across various fields.

The emergence of "augmented reality" and "engagement" as frequent keywords suggests that, alongside VR, related technologies such as augmented reality (AR) are also receiving increased amounts of attention. This may be due to AR's potential to complement VR by providing additional context or enhancing interactivity. The prominence of

**Table 6** Most Frequent Words

| No | Terms             | Frequency |
|----|-------------------|-----------|
| 1  | Students          | 18        |
| 2  | Technology        | 14        |
| 3  | Education         | 11        |
| 4  | Simulation        | 11        |
| 5  | Augmented reality | 8         |
| 6  | Engagement        | 8         |
| 7  | Environments      | 8         |
| 8  | Performance       | 8         |
| 9  | Design            | 7         |
| 10 | Science           | 7         |
| 11 | Skills            | 7         |
| 12 | Game              | 6         |
| 13 | Model             | 6         |
| 14 | Challenges        | 5         |
| 15 | Environment       | 5         |
| 16 | Impact            | 5         |
| 17 | Motivation        | 5         |
| 18 | Technologies      | 5         |
| 19 | Tool              | 5         |
| 20 | Framework         | 4         |
| 21 | Games             | 4         |
| 22 | Health            | 4         |
| 23 | Instruction       | 4         |
| 24 | Serious games     | 4         |
| 25 | Attitudes         | 3         |

"engagement" focuses on how these technologies can increase student interaction with educational materials [117].

"Performance" is a key term underscoring the importance of evaluating VR’s impact on learning outcomes, mainly whether VR helps improve student comprehension and skills. Other significant keywords, such as "design," "science," and "skills," highlight the focus on VR design, its application in the sciences, and its role in skill development. These terms illustrate the necessity of creating compelling and contextually relevant VR experiences for diverse educational settings [67].

In recent years, the increasing frequency of keywords such as “gamification” and “serious games” has led to a growing interest in using VR for serious games and gamified learning. This trend reflects the evolving approaches to integrating VR with game elements to enhance the learning experience [141]. As shown in Fig. 9, the increase in these terms over time highlights an innovative shift toward blending entertainment and education through VR, aiming to engage learners more effectively while supporting educational objectives. This evolution suggests that educators and researchers are exploring how game-based elements can leverage VR to make learning more immersive and interactive across various fields.

**3.5 Challenges and solutions (RQ4)**

The implementation of VR in education has significant potential for enhancing learning experiences but also presents several challenges. The following is a summary of the main challenges identified in the literature concerning VR use in education:



**Fig. 9** Trends in keywords over time

1) Cost and Accessibility:

One of the most frequently cited challenges in the literature is the cost of VR technology. High-quality VR hardware, such as headsets (e.g., Oculus Quest 2, HTC Vive), and accompanying equipment, such as high-performance computers, input controllers, and motion sensors, often necessitate substantial financial investment. For example, VR headsets alone can cost between \$300 and \$1000, whereas more advanced setups, including specialized computers or haptic feedback devices, may increase the cost to thousands of dollars. This upfront capital expense is often prohibitive, particularly for schools and universities in underfunded regions or developing countries [31]. Additionally, VR applications designed for educational purposes often involve licensing fees. Many platforms require subscription-based models or per-user licenses, which can impose a recurring financial burden on institutions. For institutions already operating under tight budgets, these added costs make it difficult to justify the purchase of VR technology, especially when weighing against traditional, less expensive teaching tools. This limitation is particularly pronounced in regions with economic constraints, where funding for technology-enhanced learning is minimal [39].

Accessibility is another concern closely tied to cost. Schools and universities in rural or underserved areas often lack the infrastructure to support VR. Stable internet connections, power supplies, and computer laboratories with high-performance systems are crucial for VR integration, yet they may be lacking in many regions. This issue is exacerbated in developing countries, where the digital divide restricts access to advanced technologies such as VR, leaving many institutions unable to implement these tools effectively.

The cost and accessibility issues can be addressed through several practical solutions. One key solution is for educational institutions to establish partnerships with technology companies. Companies such as Meta (formerly Facebook) or HTC, which produce VR hardware, could offer discounted or donated devices to schools and universities. In return, these companies gain access to a large market segment and can test new products in educational settings, helping reduce the initial cost of adopting VR technology. Additionally, cloud-based VR infrastructure offers a promising solution. Cloud-based VR technology allows students to use lower-spec, less expensive devices to access the same immersive experiences by shifting the computational load to

remote servers. This reduces the need for powerful computers and expensive headsets while lowering maintenance costs, as much of the system functionality is managed remotely [39].

## 2) Health and Safety Concerns.

### a) Health issues related to VR usage.

Prolonged VR exposure primarily affects the ocular and cognitive systems. The physiological demands of focusing on screens just inches from the eyes differ significantly from natural vision, which constantly adjusts to various focal distances. This sustained focus creates an artificial visual convergence–accommodation conflict, leading to eye strain, headache, and visual discomfort [132]. Traditional educational settings rarely put such continuous strain on students' visual systems, thus, VR introduces new concerns. Moreover, the mismatch between virtual and real-world distances can confuse the brain, leading to further discomfort, such as dizziness and nausea.

These issues highlight the need for adaptive technology within VR systems, such as varifocal lenses, which can dynamically adjust focal depth based on user movement and gaze [132]. The integration of eye-tracking systems in future VR devices could significantly reduce ocular strain by allowing users to engage with virtual objects more naturally and fluidly, decreasing taxation, and increasing alignment with natural vision in VR sessions. While this is a promising direction, the cost implications for educational institutions must be considered, particularly in underfunded regions. Adopting advanced, eye-friendly VR systems would likely come with additional expenses that may hinder widespread deployment in schools and universities, creating disparities in access to the technology.

Furthermore, the mismatch between physical and virtual movements leading to motion sickness poses unique challenges, particularly for students who may use VR in classrooms without proper supervision or in environments not optimized for VR use. This creates concerns not only about physical well-being but also about the quality of the learning experience. Students suffering from nausea or dizziness may find it challenging to focus, reducing the effectiveness of VR as a pedagogical tool. In this context, a controlled, incremental exposure strategy could help educators introduce VR without overwhelming users, gradually increasing session length as users adapt [62].

### b) Safety Issues During VR Usage.

The physical risks of VR extend beyond the virtual environment to real-world spaces where users interact. Unlike traditional learning environments where students are mainly stationary, VR often requires users to move, increasing the likelihood of accidents such as tripping, collisions, or falls. While these incidents may seem minor, they pose significant concerns in educational environments where liability and student safety are top priorities. Ergonomically designed spaces that account for user movement while minimizing potential hazards are crucial for safe VR deployment in schools. Moreover, the inclusion of virtual boundaries or “guardrails” within the VR system could help warn users when they are nearing real-world obstacles, although these features may further increase the cost and complexity of implementation [150].

On a psychological level, immersive VR experiences can potentially trigger emotional responses that traditional educational tools do not. This raises ethical concerns about the content that should be used in classrooms. For example, intense or realistic simulations in medical or historical contexts, while educationally valuable, could cause distress or anxiety for some students [150]. Educators must be mindful of this when selecting VR content, ensuring that simulations are appropriate for the student's age and emotional maturity. A graded content system, similar to age-based content restrictions in films or video games, could be implemented to help mitigate these risks.

### 3) Content and Quality of Experience.

The quality and relevance of VR content represent major challenges in its educational application. Poorly designed or irrelevant VR content can undermine its effectiveness as a teaching tool. Variations in content quality may stem from a lack of standardized development protocols or insufficient understanding of how to craft experiences that optimally support learning outcomes. Inconsistent or poorly structured content can lead to confusion or disengagement among students, diminishing the potential benefits of VR [105]. Additionally, content that is not regularly updated or does not reflect recent advancements in a particular discipline risk becoming obsolete, reducing its educational value.

Gender-specific considerations are also critical in VR design. Research by Makransky et al. [92] emphasizes the importance of taking gender characteristics into account when designing pedagogical agents within VR environments. This can enhance learning outcomes by helping students feel more connected and motivated, regardless of gender. Ignoring such factors could inadvertently alienate segments of the student population, thereby limiting the inclusivity and effectiveness of VR-based learning.

Developing clear guidelines and standards for creating high-quality VR content is essential to address these issues. Collaboration between developers, educators, and subject matter experts is crucial to ensure that VR experiences are engaging and pedagogically sound. Furthermore, continuous assessment and feedback mechanisms are necessary to improve VR content, ensuring its learning experiences remain relevant, up-to-date, and aligned with educational goals [9].

### 4) Integration with the Curriculum.

Effectively integrating VR into existing curricula presents a significant challenge. Without careful planning and a well-thought-out integration strategy, VR can become disconnected from core learning activities, appearing more as an additional tool than an integral part of the educational process. When VR is treated as a separate supplement, it often results in inconsistent usage or underutilization of its potential. The integration of VR into education varies significantly across educational levels, requiring distinct approaches to address unique challenges and opportunities. In primary and secondary education, the implementation of VR often faces obstacles such as limited technological infrastructure, insufficient teacher training, and the need for age-appropriate content [13]. At these levels, VR is primarily utilized to foster engagement and curiosity through interactive experiences, such as virtual field trips or simulations of natural phenomena [30, 126]. Ensuring that VR content aligns with

the developmental stages and cognitive abilities of learners is critical to its success in these settings.

In contrast, higher education, particularly within STEM disciplines, offers broader opportunities for VR applications, focusing on technical training and complex simulations (S. [81, 82]). Here, the emphasis shifts to ensuring that VR experiences meet rigorous academic standards and align with industry-specific competencies. Institutions must also address the technical demands of high-fidelity simulations and develop robust pedagogical frameworks to integrate VR effectively into specialized courses, such as engineering, medicine, and architecture. This requires careful planning to ensure that VR enhances learning outcomes rather than serving as a supplementary tool.

To support successful integration at all educational levels, curriculum designers and educators must adopt strategies that position VR as an enhancer of learning rather than a standalone solution. This involves designing lesson plans that seamlessly incorporate VR to directly support learning objectives, ensuring alignment between virtual experiences and broader instructional goals [96]. Additionally, VR should complement traditional teaching methods by reinforcing key concepts through immersive and contextually relevant learning activities. By embedding VR thoughtfully into the curriculum, educators can maximize its potential to transform teaching and learning across diverse educational contexts.

By recognizing the distinct needs of different educational levels and adopting differentiated approaches, institutions can ensure that VR integration is both developmentally appropriate and pedagogically sound—ultimately enhancing learning outcomes across diverse educational settings.

#### 5) Teacher Skills and Training.

Teachers require additional training to use VR technology effectively in educational settings [33]. A lack of skills and knowledge about VR can severely limit the potential benefits of the technology [62]. Optimally using VR requires technical knowledge about how the devices operate and a deep understanding of integrating VR effectively into teaching strategies. Without adequate training, educators may struggle to utilize advanced VR features or customize VR content to meet their needs.

Therefore, educational institutions must provide comprehensive training and support to teachers and students to ensure they are confident and competent in using VR [62]. This training should cover the technical aspects of operating VR hardware and software and pedagogical strategies for embedding VR within existing teaching practices. Practical training enhances teachers' technical abilities and helps them understand how VR can be leveraged to improve learning outcomes and foster more engaging, interactive learning environments. Providing ongoing support and opportunities for teachers to share best practices will further ensure the sustainable integration of VR into education.

#### 6) Technical and Connectivity Issues.

To function effectively, VR technology relies heavily on stable internet connections and compatible hardware. Technical issues, such as connectivity disruptions or malfunctioning devices, can hinder the learning experience and reduce the effectiveness of VR as an educational tool. VR has become particularly challenging in

regions with limited or unreliable internet access. Fast and stable internet connections are essential for streaming VR content and facilitating real-time interactions.

Additionally, hardware maintenance is crucial; VR headsets and sensors must be regularly updated and kept in good condition. Any device malfunction or breakdown can cause delays or even cancellations of learning sessions, further impacting the educational process. According to Sprenger et al. [136], preparing VR lessons can take considerable time, and technical difficulties can negatively affect students' acceptance of the technology. Therefore, ensuring that the technical infrastructure is robust and providing rapid technical support are key to minimizing disruptions and delivering a smooth learning experience. Careful planning and regular maintenance are necessary for the seamless integration of VR in education, reducing potential technical barriers that could disrupt learning.

#### 7) Social and Ethical Considerations.

VR in education raises several important social and ethical concerns that must be carefully considered. One major issue is the long-term impact of VR on students' social development. Intensive use of VR could reduce face-to-face social interactions, as students might spend more time engaging with virtual simulations than with their peers in real-life settings. This shift could negatively affect the development of crucial social skills and the ability to build healthy interpersonal relationships.

Moreover, data privacy is a critical concern. In contrast, VR technology often collects sensitive user data, including learning behaviors, physiological responses, and interactions within the virtual environment. Without proper protection, these data could be exposed or misused. It is essential to establish strict privacy policies and ensure transparency in data management to protect students' rights and prevent unauthorized access to dissemination of personal information [114].

Other ethical concerns include the potential bias in VR content and its influence on students' thinking and behavior. Thus, a thorough evaluation of VR's social and ethical impacts on education is necessary. Care must be taken to use VR technology to prioritize students' well-being and rights, ensuring that it enhances learning without compromising their social development or privacy [131].

#### 8) Resistance to Change.

Resistance to change is often a major obstacle in the implementation of new technologies such as VR in education. Both students and educators may express discomfort or skepticism toward adopting VR. For educators, concerns may include a lack of experience or training with new technologies and doubts about the effectiveness of VR in enhancing learning compared with traditional methods. They may perceive VR as insufficiently tested or incompatible with established teaching practices, making them hesitant to embrace the technology [47].

On the other hand, students might resist virtual environments because of their habitual preferences or personal discomfort. Some students may struggle to adapt to a new learning format or feel uneasy about using VR technology. Additionally, they may be skeptical about the benefits of VR, particularly if they do not immediately see improvements in their learning outcomes [127, 151].

Addressing this resistance requires effective strategies, including providing adequate training for educators, offering concrete evidence of VR effectiveness through case studies and data, and involving students in the planning and evaluation process of the

new technology. Ensuring that educators and students feel included and supported can help reduce skepticism and increase the overall acceptance of VR in educational settings. By fostering an inclusive and evidence-based approach, institutions can mitigate concerns and pave the way for smoother integration of VR in the learning process. Table 7 highlights the key challenges faced in implementing virtual reality (VR) in education, along with proposed solutions and the primary stakeholders involved. This summary provides a clear overview to assist policymakers, educators, and institutions in understanding and addressing critical issues for effective VR integration in educational settings.

3.6 Citation and collaboration analysis (RQ5)

3.6.1 Citation distribution

Global citation analysis reveals that the article by Devon Allcoat & Adrian von Mühlen (2018) leads in terms of citation count, with a total of 210 citations and an average of 30 citations per year, as shown in Table 8. This article has significantly contributed to understanding cognitive load management in VR, establishing itself as a primary reference in related research. Guido Makransky et al. [92], focusing on the application of VR in education, ranks second, with 92 citations or an average of 15.33 citations per year, highlighting the substantial impact of VR on learning effectiveness.

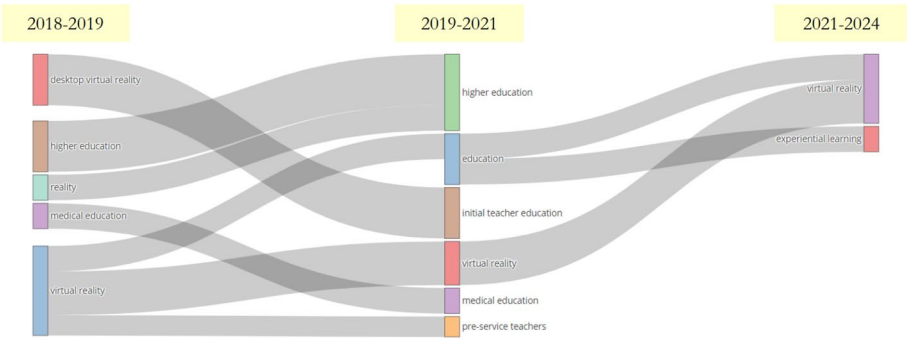
The work by Rafael Villena-Taranilla et al. [144] also shows high citation counts, with 82 citations and an annual average of 27.33, demonstrating its strong relevance and influence in recent literature. Both Marks & Thomas [95] and Jennifer Fromm et al. [47] garnered 71 and 59 citations, respectively, offering profound insights into the impact of VR on learning experiences and its integration into higher education. Older studies,

**Table 7** Challenges, proposed solutions, and stakeholders in VR implementation

| No | Challenge                         | Proposed Solutions                                                                                                                                                 | Stakeholders                                                                      |
|----|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| 1  | Cost and accessibility            | 1. Partnerships with tech companies for discounts or device donations<br>2. Use of cloud-based VR infrastructure                                                   | ■ Educational institutions<br>■ Tech companies<br>■ Government                    |
| 2  | Health and safety concerns        | 1. Development of adaptive technologies like varifocal lenses and eye-tracking<br>2. Gradual exposure strategies<br>3. Safe physical spaces and virtual boundaries | ■ VR device developers<br>■ Educational institutions<br>■ Educators and students  |
| 3  | Content and quality of experience | 1. Standardization of educational VR content<br>2. Collaboration between developers and educators<br>3. Regular content assessment and updates                     | ■ VR content developers<br>■ Educators<br>■ Educational researchers               |
| 4  | Integration with curriculum       | 1. Designing curricula with seamless VR integration<br>2. Aligning VR activities with learning objectives                                                          | ■ Curriculum designers<br>■ Educational institutions<br>■ Educators               |
| 5  | Teacher skills and training       | 1. Comprehensive training programs for teachers<br>2. Ongoing support and sharing of best practices                                                                | ■ Educational institutions<br>■ Government<br>■ Educators                         |
| 6  | Technical and connectivity issues | 1. Adequate internet and hardware infrastructure<br>2. Rapid technical support and routine maintenance                                                             | ■ Educational institutions<br>■ Internet service providers<br>■ IT support teams  |
| 7  | Social and ethical considerations | 1. Strict data privacy policies<br>2. Evaluation of content to avoid bias and negative psychological impacts                                                       | ■ Government<br>■ Educational institutions<br>■ VR content developers             |
| 8  | Resistance to change              | 1. Training and education about VR benefits<br>2. Involvement of students and teachers in technology adoption processes                                            | ■ Educational institutions<br>■ Educators<br>■ Students and education communities |

**Table 8** Distribution by Most Global Cited Documents

| No | Authors                           | Doi                           | TC  | TC per year |
|----|-----------------------------------|-------------------------------|-----|-------------|
| 1  | Allcoat & von Mühlenen [5]        | 10.25304/rlt.v26.2140         | 210 | 30.00       |
| 2  | Guido [92]                        | 10.1111/jcal.12335            | 92  | 15.33       |
| 3  | Rafael [144]                      | 10.1016/j.edurev.2022.100434  | 82  | 27.33       |
| 4  | Marks & Thomas [95]               | 10.1007/s10639-021-10653-6    | 71  | 23.67       |
| 5  | Jennifer [47]                     | 10.1016/j.jiheduc.2021.100804 | 59  | 14.75       |
| 6  | Schott & Marshall [125]           | 10.1111/jcal.12293            | 56  | 8.00        |
| 7  | Sprenger & Adrian [136]           | 10.1186/s41239-021-00243-4    | 55  | 13.75       |
| 8  | Shi [28]                          | 10.1186/s12909-020-02255-6    | 51  | 10.20       |
| 9  | González-Zamar & Abad-Segura [52] | 10.3390/educsci10090225       | 45  | 9.00        |
| 10 | Alrehaili & Al Osman [6]          | 10.1080/10494820.2019.1703008 | 39  | 13.00       |



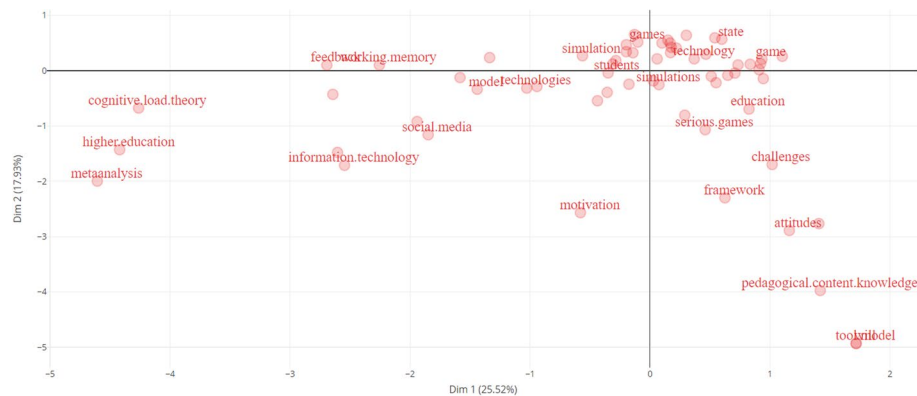
**Fig. 10** Thematic Evolution and International Collaboration in VR Research (2018–2024)

such as that of Schott & Marshall [125], include 56 citations, contributing to understanding interactive learning environments using VR. Articles such as Shi Chen et al. [28] and González-Zamar & Abad-Segura [52], with 51 and 45 citations, respectively, focus on skill development and specific applications of VR in science education. Finally, Alrehaili & Al Osman [6], with 39 citations, further advances the knowledge of interactive learning through VR. Overall, this citation distribution underscores the significant influence of research emphasizing practical VR applications, cognitive load management, and its effects on learning experiences, signaling key directions for future research and development.

3.6.2 Thematic evolution

From 2018 to 2019, research on VR in education demonstrated an initial surge in international interest, with limited collaboration predominantly involving researchers from developed countries with robust research infrastructure, as illustrated in Fig. 10. Notable contributions from Allcoat et al. [5] and Schott & Marshall [125] focused on cognitive load management and interactive learning environments. Visualizations of international collaboration networks reveal the emergence of a few key partnerships, primarily centered around established institutions in Europe and North America. These collaborations were instrumental in laying the groundwork for the development of VR technologies and their applications in education, setting the stage for future advancements and broader international participation.

From 2019 to 2021, the volume of publications and the impact of VR research significantly increased. Studies by Makransky et al. [92] and Fromm et al. [47] became



**Fig. 11** Keywords patterns

prominent, reflecting the effective utilization of VR to enhance learning outcomes and its deeper integration into educational technologies. Collaboration data during this phase highlighted a widening network, with numerous institutions from Asia and Europe joining research projects, further strengthening the global influence of these studies. Network visualizations during this time involved a growing number of international partnerships, with stronger connections between key research institutions, improving both the quality and relevance of VR research.

In the most recent period (2022–2024), international collaboration in VR research peaked, characterized by increasingly complex and integrated networks. Influential studies by Villena-Taranilla et al. [144] and Marks & Thomas [95] served as key references, demonstrating advances in VR applications for higher education and their impact on the learning experience. Visualizations of international collaboration networks depict a highly expansive web, now involving numerous countries from various parts of the world, including previously underrepresented developing nations. These collaborations have expanded the scope of VR research, increasing the quality and global influence of studies in this field and reflecting the growing trend of globalization in educational technology research.

### 3.6.3 Factorial analysis

Factorial analysis of keyword data in VR research reveals significant patterns in how core topics are classified and prioritized in the literature. The data are divided into two main dimensions (Dim1 and Dim2) and classified into one cluster, as shown in Fig. 11.

The first dimension (Dim1) underscores a significant focus on VR's technical and applicative aspects within educational settings. High positive keywords such as "technology," "augmented reality," "simulation," "engagement," and "game" indicate that the majority of related research centers on the application and design of VR technologies. This finding aligns with the growing trend of integrating gamified learning experiences into education, where VR facilitates engagement and enhances the immersive experience of learning.

Interestingly, the presence of keywords with high negative values, including "model," "impact," and "motivation," suggests a notable gap in the exploration of theoretical frameworks and the long-term effects of VR. This might indicate that while researchers are enthusiastic about the immediate applications of VR technology, comprehensive studies that assess its sustainability and effectiveness over time are lacking.

Moreover, the current emphasis on practical applications over theoretical exploration raises questions about the underlying pedagogical models employed in VR education. Future research should focus on technological advancements and consider how VR impacts learning theories and cognitive development, particularly in long-term educational settings.

Moving to the second dimension (Dim2), we observe a clearer alignment between VR and educational outcomes. Keywords such as "medical education," "performance," "experience," and "skills" are prevalent, indicating that VR is recognized for its potential to enhance learning outcomes, particularly in fields requiring practical skills, such as healthcare and engineering. This trend highlights the effectiveness of VR simulations in providing hands-on experiences that can significantly improve student performance and retention of complex concepts.

However, keywords such as "user acceptance," "feedback," and "information technology" are explored less in the literature. The lack of emphasis on user acceptance and the feedback mechanisms of VR technology suggests critical oversight. Understanding how educators and students perceive and adapt to VR technologies is essential for maximizing their effectiveness. Incorporating user-centered design principles and gathering user feedback could help refine VR applications to meet educational needs better.

Keywords such as "cognitive load theory" and "pedagogical content knowledge" have high negative values across both dimensions, indicating that VR's theoretical frameworks and pedagogical approaches remain underresearched. This gap represents a missed opportunity to connect the technological capabilities of VR with robust educational theories that could inform its design and implementation.

The interplay between technology and pedagogy is critical; understanding how cognitive load affects learning in VR environments can help design experiences that optimize user engagement and retention. Researchers can provide deeper insights into effective instructional strategies that leverage VR technology by integrating established educational theories with VR applications.

Additionally, keywords such as "higher education" and "serious games" highlight a growing interest in the application of VR within formal educational settings. This trend indicates a shift toward recognizing the potential of VR to not only enhance learning experiences but also contribute to serious gaming, which can have educational implications. Serious games can simulate real-world challenges, allowing learners to apply their knowledge in safe, controlled environments.

Overall, this analysis reveals that current VR research tends to prioritize the practical and technical aspects of the technology, with less focus on theoretical exploration and long-term impacts. This finding suggests that future studies should delve more deeply into how VR affects various educational outcomes and adoption while integrating more pedagogical theories into the research landscape.

#### **4 Discussion**

The adoption of virtual reality (VR) in education has shown significant advancements, particularly within higher education, where VR has gained traction as a tool for enhancing learning experiences. Our findings suggest that the integration of VR into education primarily aims to create immersive simulations, provide technical skill training, and foster more interactive and engaging learning environments. These results echo trends

identified in previous research, confirming the growing role of VR as a transformative educational tool [90, 118]. However, our study extends these findings by highlighting the wide-reaching impact of VR across various disciplines, including STEM and the humanities, social sciences, and healthcare.

#### 4.1 Curricular integration and theoretical frameworks

A key trend identified in our study is the increasing use of VR to simulate complex real-world scenarios, such as virtual laboratory simulations, where students can conduct high-risk experiments without needing physical facilities. This capacity to perform risky experiments in a safe environment makes VR especially valuable in fields like medicine, engineering, and chemistry, where practical training is essential for skill development. For instance, VR enables medical students to practice surgical procedures or diagnose conditions without endangering patients. This finding aligns with previous studies [47, 113], highlighting VR's potential to transform training in healthcare and other high-stakes fields. These results suggest that VR enhances practical skills and provides a safe environment for students to make mistakes and learn from them without consequences, which is critical for fields where real-world experimentation can have serious implications.

The opportunity for students to engage in real-world simulations is a significant advantage of VR, particularly in high-risk fields such as medicine and engineering. This immersive learning environment allows for experiential learning, reinforcing the importance of direct interaction with complex concepts. While VR cannot replace all aspects of hands-on learning, it complements traditional educational methods by providing students with practical experiences that might otherwise be inaccessible due to time or resource constraints.

Our study also identifies that VR benefits medical fields, where simulations are critical to developing technical skills. However, the adoption of VR is not without its challenges. For example, while VR provides a safe environment for learning high-risk procedures, the effectiveness of these simulations may be contingent on the quality of the virtual environments. This raises questions about the fidelity of the simulations and how accurately they replicate real-world conditions.

While VR is often viewed as a powerful tool for immersive learning, it is unlikely to fully replace traditional teaching methods, such as in-class learning and e-learning. Instead, VR should be considered a complementary, multimodal approach integrated into blended learning curricula. Combining classroom-based training, online learning, and VR could offer a more comprehensive and dynamic learning experience. This approach allows for specialized skill development, particularly in fields requiring hands-on experience, where VR could serve as a bridge between theoretical knowledge and practical application. Such a multimodal design enriches the learning process by addressing multiple sensory channels, thereby increasing engagement and knowledge retention [99, 112, 116].

Our study's results also emphasize that VR aligns with constructivist learning theories, in which learners build knowledge through direct interaction with their environment [79]. The immersive nature of VR allows students to engage with complex, abstract concepts in ways that traditional educational tools cannot. For example, VR simulations of medical procedures or scientific phenomena provide experiential learning opportunities,

reinforcing technical skills and abstract concepts. This finding is consistent with Kolb's theory of experiential learning, which asserts that learning occurs through direct experience [47].

The use of VR in education further supports constructivist learning principles, particularly in how students develop a deep understanding of subject matter through hands-on interaction. In fields like biology, VR simulations allow students to manipulate and observe phenomena that might otherwise be difficult or impossible to experience in a traditional classroom setting [32, 130]. As Kolb's theory emphasizes, this hands-on approach allows for deeper retention and mastery of complex subjects.

Moreover, integrating VR within educational frameworks resonates with the TPACK (Technological Pedagogical Content Knowledge) model, which underscores the interplay of technology, pedagogy, and content knowledge for effective teaching [69, 155]. VR acts as a technological affordance that enhances pedagogical strategies and content delivery simultaneously. For example, medical educators can blend VR-based surgical simulations (technology) with best pedagogical practices (guided feedback, iterative practice) and domain expertise (clinical knowledge), thus creating a cohesive multi-modal learning environment that amplifies learner outcomes.

Empirical evidence has shown that immersion in VR can significantly enhance knowledge retention and skill transfer [32, 76, 106, 130]. Studies indicate that the interactive nature of VR, which encourages active participation, improves the retention of both procedural knowledge and conceptual understanding. For example, research has demonstrated that students who engage in VR-based learning environments show greater retention of information and are better able to apply learned skills in practical, real-world scenarios [101, 108, 123]. This is particularly evident in STEM education, where VR simulations offer a more authentic and immersive experience compared to traditional methods [60, 139].

Our findings indicate that the use of VR varies significantly across educational levels, reflecting its versatility in addressing diverse learning objectives. In elementary and secondary education, VR enhances student engagement and motivation by providing interactive learning experiences. These experiences often focus on foundational concepts, such as virtual field trips to ecosystems or historical sites, enabling students to explore complex topics in an engaging and accessible manner [144]. By fostering curiosity and making abstract concepts tangible, VR supports the developmental needs of younger learners.

In contrast, higher education leverages VR for more advanced and specialized applications, particularly within STEM fields. Here, VR facilitates technical skill development through virtual laboratories and detailed 3D simulations of scientific and engineering phenomena [147]. For example, students can practice high-risk procedures, such as surgical simulations or chemical experiments, in controlled and safe environments. This allows them to gain hands-on experience and refine their technical competencies without the constraints of physical resources or safety concerns [60].

These findings underscore VR's adaptability, as it effectively caters to a wide range of educational needs, from foundational learning in K-12 settings to advanced technical training in higher education. In K-12 education, VR fosters engagement and curiosity by enabling students to explore immersive scenarios, such as traveling through the human circulatory system or witnessing historical events firsthand. Meanwhile, in

higher education, VR supports specialized learning by providing detailed simulations that align with academic and industry standards, preparing students for future professional challenges.

By addressing the unique requirements of each educational level, VR demonstrates its potential to transform teaching and learning practices, offering tailored solutions that enhance both foundational knowledge and advanced technical skills.

#### 4.2 Technical challenges and solutions

While the potential benefits of VR in education are evident, our study also identifies several significant barriers that need to be addressed for successful implementation. One major challenge is the high cost of VR hardware and software, which can limit access to this technology, particularly for educational institutions with limited budgets [31]. This issue is consistent with findings from previous studies [4], which highlight the financial barriers to widespread VR adoption in education. Additionally, the need for continuous maintenance and software updates for VR equipment, such as head-mounted displays (HMDs), adds an ongoing financial burden, complicating the integration of VR into mainstream education [115].

As highlighted by Choi et al. [31] and our findings, one of the major obstacles to widespread VR adoption is its high cost. To mitigate this, advocating for public–private partnerships is essential. Such collaborations could encourage technology companies to provide educational institutions with affordable or subsidized VR devices. These initiatives could significantly reduce financial barriers, ensuring that VR technology becomes accessible to a wider range of educational settings, especially those with limited resources.

Another challenge is the need for curriculum adjustments to integrate VR into teaching practices fully. VR could become disconnected from the core learning goals without careful alignment with educational objectives, leading to suboptimal learning outcomes [149]. As Chen et al. [28] argue, VR's effectiveness is maximized when integrated into the curriculum, complementing traditional educational methods rather than a stand-alone tool.

To overcome these challenges, we recommend several practical measures. First, educational institutions should prioritize developing curricula incorporating VR technology, particularly for simulations, technical training, and virtual laboratories [40]. For example, medical curricula could integrate VR to simulate complex surgical procedures, while engineering programs could use VR to test design prototypes in a virtual environment. These approaches would help institutions capitalize on VR's ability to provide immersive learning experiences.

In addition to curriculum development, VR developers should focus on creating more cost-effective solutions. Affordable hardware and software are essential, especially in resource-limited educational settings. By ensuring that VR devices are user-friendly and easy to maintain, developers could significantly reduce the barriers to adoption [51].

Moreover, policymakers should support VR integration by providing financial incentives, such as subsidies and grants, to help institutions adopt this technology. Partnerships between educational institutions and technology companies could also help lower costs and promote wider educational access to VR. Policymakers should also enforce

regulations to ensure the safety and quality of VR content and devices, which would help mitigate concerns about data privacy and security [153].

A critical yet often overlooked barrier is the lack of educator training in VR integration [62]. Without adequate professional development, teachers may struggle to incorporate VR into their pedagogy, limiting its educational impact [95]. Comprehensive training programs should be implemented, focusing on both technical proficiency with VR devices and pedagogical strategies to align VR with learning goals. These programs could include workshops, online certifications, and peer-learning networks to foster confidence and competence among educators.

inally, robust technological infrastructure is essential for VR's success in education. High-speed internet and reliable hardware are prerequisites for seamless VR experiences, yet many institutions, particularly in underserved areas, lack such resources [115]. Governments and institutions must invest in infrastructure development to ensure equitable access to VR, enabling all students to benefit from immersive learning opportunities.

As summarized in Table 7, addressing these challenges through strategic partnerships, curriculum alignment, affordable technology, policy support, educator training, and infrastructure investment can transform VR into a powerful educational tool, bridging gaps and enhancing learning outcomes across diverse settings.

#### 4.3 Policy implications and recommendations

At the policy level, governments should support the integration of VR in education by providing financial incentives, such as subsidies for educational institutions, grants for VR content development, or discounts on hardware for low-budget schools [56]. Moreover, funding should be allocated to establish VR infrastructure in underserved areas, ensuring every student has equitable access to immersive learning opportunities. These measures would help overcome financial and logistical barriers that currently hinder the adoption of VR in education.

For successful implementation, we recommend that schools and universities start by incorporating VR into curricula that benefit from immersive learning experiences. STEM subjects, medical training, and history education can leverage VR for simulations replicating real-world scenarios, such as conducting virtual chemistry or biology experiments and practicing surgical procedures. To overcome the high costs of VR technology, educational institutions should foster partnerships with technology companies to obtain affordable VR hardware and software. Furthermore, professional development programs for educators should be prioritized to ensure they are equipped to integrate VR tools effectively into their teaching strategies [61].

The adoption of VR offers substantial benefits for educators and institutions. For teachers, VR enables immersive, engaging experiences that enhance student motivation and improve retention of complex concepts by making abstract ideas tangible and personalized. For institutions, particularly in higher education, integrating VR positions them as leaders in educational innovation, attracting prospective students seeking advanced, technology-driven learning environments [143]. By embracing VR, institutions can prepare students for future careers in a technology-centric world while enhancing their reputation for forward-thinking education.

In summary, strategic policy interventions—financial incentives, infrastructure investment, curriculum integration, industry partnerships, and robust educator training—can overcome barriers to VR adoption. These steps will position VR as a transformative tool, fostering equitable, innovative, and impactful education across diverse settings.

## 5 Conclusion

This study contributes to the growing body of literature on immersive technologies by offering a comprehensive bibliometric analysis of virtual reality (VR) applications in education. By examining 97 peer-reviewed articles published between 2018 and 2024, the study identified key publication trends, disciplinary distributions, citation patterns, and research gaps. The findings underscore the potential of VR to transform educational practices, particularly through interactive simulations, personalized learning environments, and hands-on, experiential training.

While the results confirm a steady increase in scholarly attention toward VR, several critical research gaps remain. There is limited evidence on the long-term cognitive, emotional, and developmental impacts of VR-based instruction, especially among young learners. Further, the potential for bias embedded in VR content and its implications for equity and learning outcomes warrant systematic investigation. Although VR has demonstrated effectiveness in developing technical competencies, its role in fostering soft skills, such as leadership, collaboration, and emotional intelligence, remains underexplored.

Methodologically, the study is constrained by the scope of the databases and analytical tools employed. While bibliometric approaches are well-suited for mapping large-scale research trends, they may not capture the lived experiences, contextual barriers, or pedagogical nuances associated with the implementation of VR. Thus, future research should adopt qualitative or mixed-methods designs to investigate user perceptions, educator preparedness, and institutional readiness for integrating VR.

Despite these limitations, the study affirms the growing influence of VR in education and its capacity to enrich learning through immersive and student-centered approaches. However, several challenges must be addressed to enable widespread adoption, including high implementation costs, infrastructure disparities, curricular integration, and professional development for educators. Advancing the integration of VR in education will require sustained collaboration among policymakers, educators, researchers, and technology developers. By addressing these structural and pedagogical challenges and expanding the methodological scope of future research, the educational potential of VR can be more fully realized across diverse learning environments.

## 6 Limitations and future work

### 6.1 Limitations

This study is subject to several limitations, primarily related to database selection, publication coverage, and methodological scope. The analysis relied exclusively on three major Web of Science indices—the Science Citation Index Expanded (SCIE), Social Sciences Citation Index (SSCI), and Emerging Sources Citation Index (ESCI)—as well as on a subset of widely recognized publishers. While these sources are reputable and widely used in bibliometric research, their scope may exclude relevant publications from smaller or regionally focused journals not indexed in these databases. Consequently,

potentially valuable insights from underrepresented regions or less mainstream publishers may have been omitted.

Temporal limitations also apply, as the analysis focused on articles published between January 2010 and August 2024. While this timeframe captures recent trends, it may not fully reflect earlier foundational work or emerging research published after the data collection window. Additionally, the methodological approach—particularly bibliometric analysis and network visualization—is inherently quantitative and may not capture the literature's more nuanced theoretical or pedagogical aspects. These techniques provide valuable macro-level insights but may overlook subtle dimensions such as cultural context, learner experience, or instructional design theory.

## 6.2 Future work

Future research on VR in education would benefit from a more integrated, interdisciplinary, and longitudinal approach. A key direction involves exploring the convergence of artificial intelligence (AI) and VR. AI can enable adaptive learning systems that personalize content delivery based on students' progress, preferences, and learning styles [120]. Machine learning algorithms can dynamically tailor feedback, pacing, and task complexity to optimize individual learning outcomes.

Enhancing interactivity and immersion is another important focus. The integration of gamification elements, interactive simulations, and hands-on tasks can enrich learner engagement and facilitate skill acquisition in more dynamic and meaningful ways [119, 121]. As hardware continues to evolve—offering higher resolution, lower latency, and improved haptic feedback—future VR platforms will support increasingly realistic and pedagogically practical learning experiences. Additionally, longitudinal studies are necessary to assess the long-term cognitive, social, and emotional effects of VR-based instruction. These studies could provide a deeper understanding of how immersive environments influence student motivation, collaboration, and self-regulated learning across diverse educational contexts.

Addressing the affordability and accessibility of VR technologies remains a crucial challenge. Future work should investigate scalable solutions that facilitate wider adoption across diverse socioeconomic backgrounds, particularly in underserved communities. Integrating VR with collaborative, project-based, and experiential learning models could further enhance its educational potential. Ultimately, developing robust evaluation frameworks is crucial for assessing the effectiveness of VR interventions. Future studies should focus on defining valid learning metrics and identifying pedagogically grounded methods for evaluating engagement, skill transfer, and learning outcomes [122]. Such frameworks will help ensure that VR remains innovative and has an educational impact.

## 6.3 Posted Preprint Statement

This article was previously published as a preprint on Research Square on September 9, 2024 (Version 1) and September 30, 2024 (Version 2). The latest version of the preprint can be accessed through the following link: <https://doi.org/https://doi.org/10.21203/rs.3.rs-5051188/v2>. This work has not been published in any journal and has not undergone peer review. We affirm that we will comply with the journal's policies regarding preprints throughout the submission and review process.

### Author contributions

The idea for the article was conceived by A.D.S. The literature search and data analysis using bibliometric methods following PRISMA guidelines were performed by A.D.S., S.R., S.R.D., and R.F. Data curation was carried out by A.D.S., R.P.S., and R.F. The manuscript was originally drafted by A.D.S. and critically revised by S.C. and R.P.S. The visualization work was done by A.D.S. All authors commented on previous versions of the manuscript. A.D.S. supervised the overall work. All authors read and approved the final manuscript.

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### Data availability

The data used in this research is available upon request. Interested parties can access the dataset by contacting the corresponding author. The dataset includes all relevant documents and information analyzed during the study, ensuring transparency and reproducibility of the research findings. The dataset and appendix can be accessed at the following link: [https://osf.io/rd4jn/?view\\_only=1eef2b49ce044aa3bc06289c394af8e4](https://osf.io/rd4jn/?view_only=1eef2b49ce044aa3bc06289c394af8e4).

### Declarations

#### Ethics approval and consent to participate

Ethical approval was not required for this study as it involved the analysis of publicly available bibliometric data and did not include any human or animal subjects.

#### Declaration of generative AI and AI-assisted technologies in the writing process

While preparing this work, the author(s) used OpenAI's ChatGPT and Grammarly to improve the readability and language of the manuscript. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

#### Statement regarding research involving human participants and/or animals

This research did not involve any human participants or animals.

#### Consent for publication

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#### Competing interests

The authors declare no competing interests.

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